

Augur: an α -domain variable-projection DOAS retrieval, and what its diagnostics reveal that standard fit reports do not

First Author¹ and Co Author¹

¹Gwangju Institute of Science and Technology (GIST), Gwangju, Republic of Korea

Correspondence: First Author (email-to-be-added)

Abstract. A DOAS fit reports a residual, a chi-square, an uncertainty and a convergence flag, and these are read as a description of how well the retrieval did. Each can assert more than the fit knows, and the gap is invisible from inside the fit report. We present Augur, an open retrieval for broadband absorption spectra that is configured by instrument and campaign profiles rather than written for one instrument, and use one field case to measure that gap: 53 days of measurements in 2026 with a thermal-dissociation broadband cavity-enhanced absorption spectrometer with two heated and one ambient-temperature cavity, measuring NO₂, total peroxy nitrates and total alkyl nitrates (Min et al., 2016; Nam et al., 2022).

Augur poses the retrieval in the extinction domain as a separable nonlinear least-squares problem and solves it by variable projection with an analytic two-term Golub–Pereyra Jacobian, propagating a joint covariance that includes the nonlinear parameters. Its contribution, however, is architectural rather than numerical. The time-resolved references enter as data, so a leave-one-out perturbation of the calibration record is a loop over knots rather than a reprocessing campaign; and the projected objective is available as a function of the nonlinear parameters, so the shape of the minimum can be mapped rather than inferred from a converged point. On synthetic spectra an independent DOAS code reproduces Augur to within its output resolution; on the field record the two agree in NO₂ with $r^2 \geq 0.990$ but differ in scale by up to 6 % (Sect. 3.3). Neither of these questions can be put to a conventional implementation without reprocessing the campaign once per answer.

Asking them changes what the uncertainty means. For the product the instrument reports, formed as the difference of two retrievals, leave-one-out perturbation of the zero-air, reflectivity and etalon references in the best-calibrated week of the deployment gives a structural uncertainty 1.3 to 2.3 times the sigma the fit reports for the same 60 s records, so the total is 1.6 to 2.5 times what is reported. The uncertainty column delivered with the product is not that quantity at all but a gain systematic proportional to NO₂, and carries none of this. On synthetic spectra with known truth the realised scatter is 0.91 to 1.32 times the reported sigma under white noise (within 10 % in 15 of 30 cases; 1.22 to 1.32 in 9 of 13 synthetic instruments at nominal noise, where the free wavelength shift adds scatter the covariance does not carry) and 1.42 to 2.05 times under autocorrelated noise. None of the calibration-chain terms enlarges the residual.

Two failures are of a different kind, because the information existed upstream and was discarded. Wavelength shift is identifiable in one window and not in another – the profiled objective of twelve ambient spectra varies by a median 1.2 % over ± 4 px where the other window spans five orders of magnitude – yet both are reported as converged, and under matched settings 85.5 % of them terminate at a parameter bound; a laboratory spectrum with added NO₂ locates the same shift to a tenth of a pixel, two pixels from where the ambient fit converged. And a nine-hour misalignment between the reflectivity time base and

the spectra, present in twelve days of the operational product, displaced the retrieved alkyl nitrates by 0.100 ppb, 1.44 times the entire structural budget, while moving the fitted wavelength shift by 0.005 px and changing the residual and the reported uncertainty by two parts in a thousand.

Neither failure is expensive to prevent: the termination state, the distance to the nearest reference in time and the provenance of each reference are all known inside the retrieval and need only be carried to the product, which we recommend as a reporting convention. Augur exports the termination state and the distance to the nearest reference and records which reference file was used; recording each reference's build time and time convention is specified but not yet implemented. We release Augur and a synthetic benchmark with which any DOAS implementation can measure the same quantities.

1 Introduction

A DOAS retrieval returns more than a concentration. It returns a residual, a chi-square, a parameter covariance and a convergence flag, and these travel with the concentration into averaging, into filtering, into trend fits and into comparisons with models and with other instruments. In practice they are read as a statement about how much the retrieval can be trusted: a small residual means the model described the spectrum, a converged flag means the optimiser found the answer, and the reported 1σ means the concentration is known that well. Each of these readings is conditional, and the conditions are not reported alongside the numbers.

That the conditions matter is not a new observation. The uncertainty a DOAS fit reports is the propagation of the residual scatter through the linear system, and it is correct only if the residual is white and the forward model is complete; Stutz and Platt (1996) showed that structure in the residual inflates the true error well beyond the reported one, and the standard treatment (Platt and Stutz, 2008) sets out the conditions explicitly. Practitioners respond with rules of thumb – multiply the reported error by two, or by the ratio of the residual to the photon noise – and with sensitivity studies on the spectral fit: the choice of window, the polynomial order, the set of cross sections included.

What that literature examines is the spectral fit. The quantities the fit treats as given – the reference spectrum, the mirror reflectivity, the instrument line shape – are usually characterised separately, assigned a calibration uncertainty, and combined with the fit uncertainty at the end. For a cavity-enhanced instrument this is a stronger assumption than it appears. The reference and the reflectivity are not constants with an uncertainty; they are time series, measured intermittently and interpolated to every measurement, and the error they contribute to a given record depends on where that record sits relative to the nearest calibration. A campaign-mean calibration uncertainty, of the kind reported for the effective path length in earlier instruments of the design used here (Min et al., 2016; Nam et al., 2022), does not describe it.

Intercomparison campaigns approach the question from the other side. By running many implementations on the same spectra – as CINDI-2 (Kreher et al., 2020) and its predecessors have done for MAX-DOAS – they measure the spread between codes, which bounds implementation error and has been decisive in standardising settings. But a spread is not an error: when implementations agree they may agree on a common mistake, and when they disagree the comparison cannot say which is right, because there is no truth. Synthetic studies supply the truth that intercomparisons lack, and are the standard way to verify

a solver, but they are normally built to test the fit – given a spectrum, is the concentration recovered – and so inherit the same boundary: the references are correct by construction.

This paper takes the boundary as the subject. Treating the calibration chain as part of the retrieval rather than as its input divides the ways a fit report overstates its knowledge into two kinds, which want different remedies. The first is what the fit cannot know: contributions that leave the residual unchanged, so that no amount of care in reading the fit report will reveal them. These have to be measured by perturbing the retrieval and re-running it, and added to the budget. The second is what the pipeline already knows and does not carry to the product: whether the optimiser stopped at a bound, whether the nearest reflectivity measurement was fifteen minutes or eleven hours away, whether a quality gate upstream rejected the calibration blocks on either side. This information exists inside the software at the moment the concentration is computed and is discarded between one step and the next. It costs nothing to keep, and keeping it turns a silent failure into a flagged one.

We present Augur, an open retrieval built so that both kinds of question can be asked cheaply enough to ask routinely. The retrieval is posed in the extinction domain as a separable nonlinear least-squares problem and solved by variable projection. It is not written for one instrument: the layout of the raw data, the calibration flags and the fit configuration are read from instrument and campaign profiles (Appendix A), and the same code runs other configurations through such profiles. The diagnostics are demonstrated here on one field case, a thermal-dissociation broadband cavity-enhanced absorption spectrometer with two heated and one ambient-temperature cavity measuring NO_2 , total peroxy nitrates and total alkyl nitrates over 53 days in 2026; other configurations are not evaluated in this paper. We verify it against synthetic truth and against an independent DOAS implementation, measure the structural terms it cannot report – which together exceed the sigma it reports by a factor of 1.6 to 2.5 in the best-calibrated week – map the identifiability of its nonlinear parameters, and describe twelve days of operational product built from a calibration reference that a later correction had made stale, an error 1.44 times that budget which changed the residual and the reported uncertainty by two parts in a thousand while moving the 300 °C channel’s concentration by two reported standard deviations. We propose a small reporting convention whose distance fields would have flagged it at the first reprocessing and whose provenance fields, once implemented, would name its cause, and release a synthetic benchmark with which any DOAS implementation can measure the same quantities on the same cases.

The evidence is one field case – one instrument over one campaign – which bounds what can be concluded about instruments in general. Two things are intended to carry beyond it. The benchmark of Sect. 3.4 is built from cross sections and a wavelength calibration but contains no retrieval code and no field data, so the measurements it supports do not depend on any retrieval implementation, Augur included; its cross sections, fitting windows and wavelength grid are those of one instrument, and carrying it to another requires that instrument’s own. And the reporting convention of Sect. 6 asks for quantities that every implementation already computes, apart from the build time and time convention of each reference, which must be recorded when the reference is built; adopting it requires no change of algorithm. Where a result does depend on this instrument – the magnitude of the reflectivity term, the behaviour of the cold channel – we say so.

Figure 1 previews the argument: what a converged fit reports, against what the data support. Section 2 sets out the retrieval and what the implementation exposes. Section 3 verifies it. Sections 4 and 5 take the two kinds of failure in turn, and Section 6 draws the reporting convention out of them and names what it cannot reach. Section 7 demonstrates the retrieval on the field

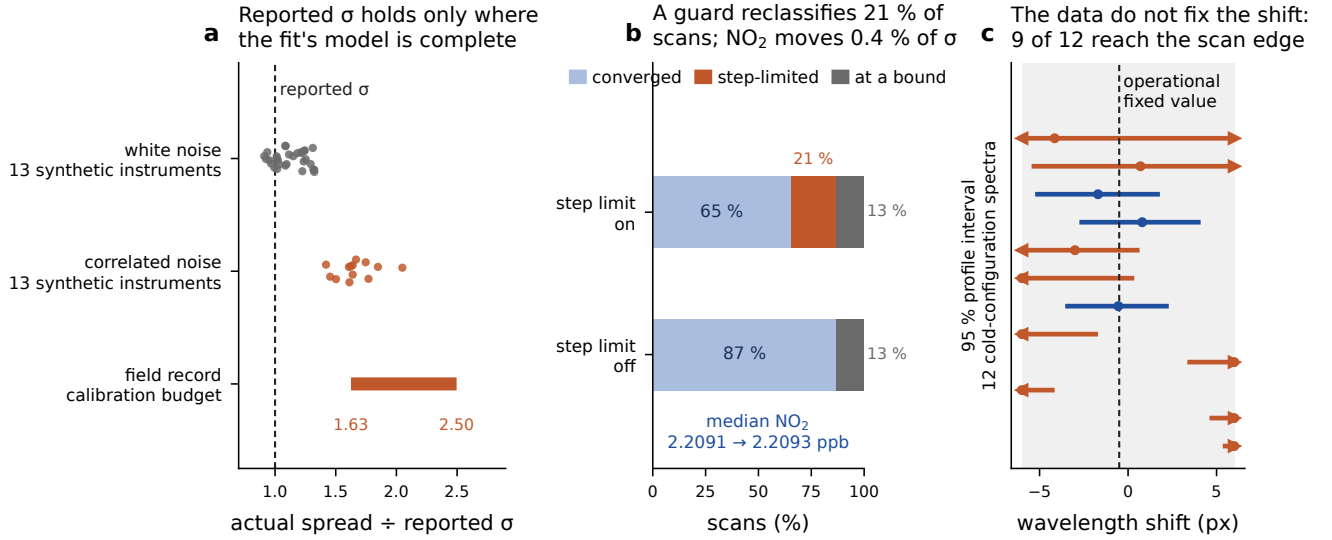


Figure 1. What a converged fit reports, and what the data support. **(a)** Spread of the retrieved NO₂ divided by the uncertainty the fit reports. Grey points: white noise on synthetic spectra, 30 cases over 13 synthetic instruments spanning line-shape widths of 0.5–2 times nominal, windows of 0.7–1.3 times, pixel dispersions of 0.7–1.4 times, signal-to-noise over a factor of 32 and fitted polynomial orders 2–6. Orange points: the same instruments with correlated (AR(1), $\rho = 0.5$) noise. Bar: the field record, for which the calibration-chain budget of Sect. 4 gives a total uncertainty 1.63 (robust scale) to 2.50 (standard deviation) times the fit-propagated σ ($n = 8847$ records of 60 s). **(b)** Termination state of 8446 operational scans with the optimiser’s step limit on, as operated, and off; the median retrieved NO₂ of the same scans is given below the bars. **(c)** 95 % profile-likelihood intervals of the wavelength shift for 12 spectra in the cold configuration, from the profiled objective scanned over ± 6 px (shaded), with $\text{RSS}/\text{RSS}_{\min} \leq 1 + F_{0.95}(1, n - P)/(n - P)$ and $n - P = 766$. Dots mark the minimum; arrows mark intervals that reach the edge of the scan. Dashed: the value at which operation fixes the shift, -0.5 px.

case, and Sect. 8 concludes. Appendix A collects the instrument and retrieval configuration, Appendix B the projected Jacobian and joint covariance, and Appendix C the measurements on calibration-block length.

2 Retrieval formulation

2.1 Forward model in the extinction domain

100 The cavity output is converted to an extinction coefficient before any spectral fitting is attempted. For a sample spectrum $I(\lambda)$ and a reference spectrum $I_0(\lambda)$ recorded with the cavity filled by zero air,

$$\alpha(\lambda) = \underbrace{\left[\frac{R_L(1-R(\lambda, t))}{d} + R_L \alpha_{\text{Ray}}(\lambda; T_0, P_0) \right]}_{\alpha_{\text{mirror}}} \frac{I_0(\lambda) - I(\lambda)}{I(\lambda)} - [\alpha_{\text{Ray}}(\lambda; T, P) - \alpha_{\text{Ray}}(\lambda; T_0, P_0)], \quad (1)$$

where $R(\lambda, t)$ is the mirror reflectivity, d the mirror separation, R_L the ratio of the full mirror separation to the effective cavity length actually filled by the sample (i.e. excluding the purge-gas volumes at the mirrors), and α_{Ray} the Rayleigh extinction evaluated at the temperature and pressure of the sample (T, P) and of the reference (T_0, P_0) . The second bracket removes the Rayleigh difference between the two states; the first converts a relative intensity change into an absolute extinction.

Working in α rather than in optical density has one consequence that organises everything below: **the gas amounts enter Eq. (1) linearly**. The quantity to be fitted is a sum of reference cross sections with concentration coefficients, plus a smooth background, plus whatever periodic structure the optics impose. No logarithm stands between the unknowns and the measurement.

2.2 Separable structure

Let p index the n detector pixels of the fitting window and let

$$x(p) = \frac{2(p - p_{\min})}{p_{\max} - p_{\min}} - 1 \in [-1, 1].$$

The model for the measured extinction is

$$\alpha_{\text{model}}(p) = \sum_{i=1}^m c_i \tau_i(T) \sigma_i(p'_i) + \sum_{j=0}^q b_j T_j(x(p)) + a_s \sin(fp) + a_c \cos(fp), \quad (2)$$

$$p'_i = (p - p_c) s_i + p_c + \delta_i, \quad (3)$$

with T_j the Chebyshev polynomial of the first kind, p_c a fixed window centre, and $\tau_i(T) = 1 + \kappa_i(T - T_{\text{ref}})/100$ the temperature scaling of the i -th cross section. Each reference carries its own wavelength shift δ_i (pixels) and squeeze s_i ; in practice one gas is designated the anchor and the companions are linked to it or fixed, which is a constraint on θ and not a change of structure.

Two choices in Eq. (2) deserve comment.

The background is Chebyshev, not monomial. On $x \in [-1, 1]$ the Chebyshev basis is far better conditioned than $1, p, p^2, \dots$; with the monomial basis the polynomial block alone reaches condition numbers that make the background coefficients meaningless, and the ill-conditioning leaks into the gas coefficients through the shared normal equations.

The etalon enters as a sine-cosine pair at fixed frequency, not as an amplitude-phase pair. Since

$$A \sin(fp + \varphi) = A \cos \varphi \sin(fp) + A \sin \varphi \cos(fp),$$

the two parametrisations span the same model space, but the pair (a_s, a_c) is **linear** whereas (A, φ) is not. Writing it this way removes a nonlinear parameter from the search and makes the etalon amplitude and phase part of the globally solvable linear step. Only the frequency f remains fixed a priori.

Collecting the $k = m + (q + 1) + 2$ basis vectors into a design matrix $\mathbf{A}(\theta)$ whose columns depend on the nonlinear vector $\theta = (\delta_i, s_i)$ only through Eq. (3), and writing $\mathbf{W} = \text{diag}(w_1, \dots, w_n)$ for the pixel weights, the problem is

$$\min_{\theta, \mathbf{c}} \|\mathbf{W}(\mathbf{y} - \mathbf{A}(\theta) \mathbf{c})\|_2^2, \quad \mathbf{y} = \alpha_{\text{meas}}. \quad (4)$$

This is a **separable** nonlinear least-squares problem in the sense of Golub and Pereyra (1973): k linear parameters and, for the operational configuration, two to six nonlinear ones.

135 2.3 Variable projection with an analytic projected Jacobian

For any θ the inner problem is an ordinary linear least-squares problem, so $\mathbf{c}$ need never be searched. Substituting the minimiser $\hat{\mathbf{c}}(\theta) = \mathbf{A}^+ \mathbf{y}$ leaves the *projected* residual

$$\mathbf{r}(\theta) = (\mathbf{I} - \mathbf{A}(\theta)\mathbf{A}(\theta)^+) \mathbf{y} = \mathbf{P}_{\mathbf{A}(\theta)}^\perp \mathbf{y}, \quad (5)$$

a function of θ alone. We minimise Eq. (5) with a bounded Levenberg–Marquardt/trust-region solver. In implementation the weighted, column-scaled design matrix is factorised as $\mathbf{AS}^{-1} = \mathbf{QR}$, which delivers $\hat{\mathbf{c}}$, the residual, and the projector $\mathbf{P}^\perp = \mathbf{I} - \mathbf{QQ}^\top$ from one factorisation.

The Jacobian of Eq. (5) is supplied analytically in the full Golub–Pereyra form,

$$\frac{\partial \mathbf{r}}{\partial \theta_k} = - \left[\underbrace{\mathbf{P}^\perp \mathbf{D}_k \hat{\mathbf{c}}}_{\text{projection term}} + \underbrace{(\mathbf{A}^+)^{\top} \mathbf{D}_k^{\top} \mathbf{r}}_{\text{second term}} \right], \quad \mathbf{D}_k = \frac{\partial \mathbf{A}}{\partial \theta_k}, \quad (6)$$

with $(\mathbf{A}^+)^{\top} = \mathbf{QR}^{-\top}$. Only the gas columns of $\mathbf{D}_k$ are non-zero: differentiating Eq. (3),

$$145 \quad \frac{\partial}{\partial \delta_i} \sigma_i(p'_i) = \sigma'_i(p'_i), \quad \frac{\partial}{\partial s_i} \sigma_i(p'_i) = (p - p_c) \sigma'_i(p'_i), \quad (7)$$

so σ'_i is obtained once from the derivative of the interpolating spline that already represents each reference. The polynomial and etalon columns are independent of θ and contribute nothing. Both terms of Eq. (6) cost one triangular solve and one multiplication by $\mathbf{Q}$; the derivation and the invariance of Eq. (6) under the column scaling $\mathbf{S}$ are given in Appendix B.

Retaining the second term makes Eq. (6) the exact derivative rather than the Kaufman (1975) approximation. The gradient of the objective is the same under both, because $\mathbf{A}^+ \mathbf{r} = 0$; what the second term adds is the exact curvature and, because the derivative is exact, a check against finite differences to the level of round-off, which turns the Jacobian into a regression test rather than a claim (Sect. 3.2). We have not compared convergence rates of the two forms.

Relation to existing DOAS implementations. The separable structure is not new to DOAS; what is absent from the established treatments is the derivative of the linear solution with respect to the nonlinear parameters. QDOAS solves the slant columns and the polynomial by singular value decomposition inside each evaluation of the fitting function while its Marquardt–Levenberg loop iterates over the nonlinear parameters only – the structure WinDOAS itself named “Marquardt-SVD” – but its source states that the derivatives with respect to shift and stretch are “always numeric”, so no term of the form of Eq. (6) is formed. DOASIS (Kraus, 2006) goes the other way: once shift and squeeze are fitted, the concentrations are handed to the nonlinear solver as well, and the thesis notes that the algorithm may then fail to find the global optimum that the linear step would have guaranteed. The standard textbook scheme (Platt and Stutz, 2008, Sect. 8.3.4) alternates one linear solve with one Levenberg–Marquardt step, explicitly “with fixed parameters a_j and c_h ” – block-coordinate optimisation, in which $\partial \hat{\mathbf{c}} / \partial \theta$ does not appear. Outside DOAS, the analytic form of Eq. (6) is established in atmospheric remote sensing: Bärligea et al. (2023)

use exactly this projected derivative, with the Kaufman variant, for a SWIR retrieval in which – as in BIRRA (Gimeno García et al., 2011) – the separability is inverted relative to Eq. (2).

165 2.4 Time-resolved mirror reflectivity

R in Eq. (1) is not a constant of the deployment. It is determined at intervals from helium injections and drifts between them. Rather than assigning one value per campaign, or per day, we carry $R(\lambda, t)$ as a monotone piecewise-cubic (PCHIP) interpolation through the injection knots, evaluated at the timestamp of each spectrum. Monotone interpolation is chosen so that the reconstruction cannot overshoot between knots; the cost, quantified in Sect. 4.4, is that it also cannot represent a turning point that falls between two knots.

2.5 Uncertainty

With $\hat{\sigma}^2 = \mathbf{r}^\top \mathbf{r} / (n - P)$, where P counts the linear columns and the active nonlinear parameters, the conditional covariance of the linear coefficients is the usual

$$\text{Cov}(\mathbf{c} \mid \theta = \hat{\theta}) = \hat{\sigma}^2 (\mathbf{A}_{\mathbf{W}}^\top \mathbf{A}_{\mathbf{W}})^{-1}. \quad (8)$$

175 Equation (8) is what DOAS retrievals conventionally report, and it is conditional: it holds the fitted shift and squeeze at their optimum. Variable projection separates θ from $\mathbf{c}$ in the *solution*, but it does not separate them in the *uncertainty* – if the shift is uncertain, the concentrations inherit that uncertainty. We therefore also form the joint covariance from the full Jacobian of the unprofiled residual,

$$\mathbf{J} = [\mathbf{A}_{\mathbf{W}} \mid \mathbf{V}], \quad \mathbf{V}_{:,k} = \sum_i \frac{\partial \mathbf{A}_{:,i}}{\partial \theta_k} \hat{c}_i, \quad \text{Cov}(\mathbf{c}, \theta) = \hat{\sigma}^2 (\mathbf{J}^\top \mathbf{J})^{-1}, \quad (9)$$

180 and take the gas block (O’Leary and Rust, 2013). Because the inverse of the gas block of Eq. (9) is the Schur complement $\mathbf{A}_{\mathbf{W}}^\top \mathbf{A}_{\mathbf{W}} - \mathbf{A}_{\mathbf{W}}^\top \mathbf{V} (\mathbf{V}^\top \mathbf{V})^{-1} \mathbf{V}^\top \mathbf{A}_{\mathbf{W}}$, which is never larger than the matrix inverted in Eq. (8), the joint standard errors are never smaller than the conditional ones. Note that $\mathbf{V}$ holds $\mathbf{c}$ fixed and is therefore *not* the projected Jacobian of Eq. (6); using Eq. (6) here would count the linear block twice. Both quantities are reported, never one overwriting the other, so that the size of the difference can be read directly (Appendix B4 gives it for a small test problem).

185 When $n - P \leq 0$ the variance estimate is not defined and is reported as missing rather than as a finite number; the scan is flagged as underdetermined and retained.

2.6 Termination state

A fit that stops because it hit a parameter bound, and a fit that stops because the gradient vanished, are not the same result, but a residual-and- χ^2 report cannot tell them apart. We therefore classify and export the termination state of the nonlinear stage separately from the solver’s own success flag: CONVERGED, AT_BOUND (a nonlinear parameter rests on its box constraint at exit), STEP_LIMITED (the trust-region step was capped), MAX_NFEV, and FAILED. The classifier is deliberately conservative by construction: a nonlinear parameter within a small tolerance of its box constraint (relative to the box width) is flagged

even if the optimum happens to lie there, and across the solves of one record the least favourable state is kept. It changes no retrieved value. Its effect is on what the record admits, and that effect is large (Sects. 5.2 and 7.3).

195 2.7 Implementation note: scaling of the fitted quantity

Equation (4) is solved on α scaled to order unity, and the linear coefficients are scaled back afterwards. This is not cosmetic. Ambient α in these windows is of order 10^{-7} cm^{-1} , so residuals and gradients are of order 10^{-15} , while the default termination tolerances of a general-purpose least-squares routine are of order 10^{-8} . On unscaled α the termination test can then be satisfied at the starting point: the solver returns after a single function evaluation, having moved the shift by nothing, and reports success.

200 In one operational spectrum the unscaled fit exited with shift $+0.0000 \text{ px}$ after $n_{\text{fev}} = 1$, while the same spectrum scaled by 10^7 converged to -0.5000 px after $n_{\text{fev}} = 5$. We note this because it is silent – the failure mode produces a plausible number with a success flag – and because it is specific to re-implementations in scaled, tolerance-driven optimisers rather than to the DOAS method itself.

2.8 Constraints and options not used operationally

205 For completeness: the linear step can be solved with a non-negativity constraint on the gas coefficients, and a Tikhonov penalty can be applied to the gas, etalon and auxiliary columns (never to the Chebyshev background, which would distort it). Neither is used in the configuration reported here – the operational λ is zero and the concentrations are left unconstrained so that values near zero remain unbiased. The analytic Jacobian of Eq. (6) requires the unconstrained linear step: at an active non-negativity bound the projector is not differentiable, and the implementation falls back to finite differences in that case. Robust (Tukey
210 bisquare) iteratively reweighted fitting is available and likewise unused here.

2.9 What the implementation exposes

The formulation of Sects. 2.1–2.8 is not unusual, and Sect. 3.3 shows that a conventional implementation solving the same problem returns the same concentrations on synthetic spectra, and on the field record NO_2 amounts that agree closely but not exactly in scale. What differs is what can be asked of the retrieval afterwards, and the difference is one of cost rather than of
215 possibility.

Two questions are asked repeatedly in this paper. The first is what happens to the product when a calibration reference is changed: the structural budget of Sect. 4 removes one zero-air knot, one reflectivity knot or one etalon component at a time and re-runs the retrieval over the 9034 records of the budget window that have a reflectivity knot within 1 h (8847 after bound-terminated fits are excluded), once per knot and then for the pairwise and triple combinations. The second is what the objective
220 looks like away from the converged point: the identifiability map of Sect. 5 evaluates the projected objective on a grid of 160 points in the wavelength shift for each of twelve spectra, and repeats a coarser version over 165. Neither question is answered by a fit report; both are answered by re-entering the retrieval with one input altered.

Table 1. Retrieval configurations used in Sects. 3–6, in short form; windows are rounded to the nanometre, and the full configuration, including the exact window bounds, is in Table A1.

Label	Cell	Window (nm)	Baseline	Shift range (px)	Light level
hot ANs	300 °C	430–462	4th order	−10 to 0.5	nominal
hot PNs	180 °C	444–471	3rd order	−5 to 5	nominal
cold	ambient	438–476	4th order	fixed at −0.5	reduced

Augur is arranged so that both are loops rather than reprocessing campaigns. The time-resolved references enter the forward model as data – an array of knots with times, interpolated at evaluation – rather than as a configured input file, so substituting one is an assignment. The projected objective is a callable of the nonlinear parameters, separate from the optimiser that searches it, so evaluating it at a chosen shift does not require converging a fit. In an implementation where the reference is a file named in a configuration and the objective is reachable only through the optimiser, the same two questions cost one reprocessing of the campaign per answer: several thousand for the budget, one hundred and sixty per spectrum for the map.

This is the sense in which the contribution of the software is architectural. It is not that the quantities in Sects. 4 and 5 cannot be obtained elsewhere. It is that obtaining them has to be cheap enough to do by default, because the error of Sect. 5.4 – a nine-hour error in a reference time base, invisible in the residual – could be measured only by re-running the retrieval with the time base changed.

3 Verification

Five things are verified, and they are different things. That the implementation solves the equations of Sect. 2; that the analytic Jacobian is the derivative it claims to be; that an independent implementation of the same problem returns the same answer; that the diagnostics of Sects. 4 and 5 measure what they are said to measure, on data whose truth is known; and that the two hot channels share a relative scale. Absolute concentration accuracy against a reference standard is not established here and is not claimed anywhere in this paper.

Results in Sects. 3–6 are reported for three retrieval configurations taken from the field case of Sect. 7 and labelled by the cells they come from. What distinguishes them for the retrieval is listed in Table 1: the fitting window, the order of the baseline polynomial, the permitted shift range and the light level; the full configuration is in Appendix A. None of the results in Sects. 3–6 depends on the chemistry the cells measure; where a result depends on a property of a configuration, that property is named. Figure 2 shows one record retrieved in all three configurations.

3.1 Recovery of a known truth

Spectra are generated from the forward model of Sect. 2.1 with prescribed gas amounts, a Chebyshev baseline, a wavelength shift and a squeeze, and retrieved with the same model. Over 27 combinations – nine shifts from −4 to +4 px (including sub-

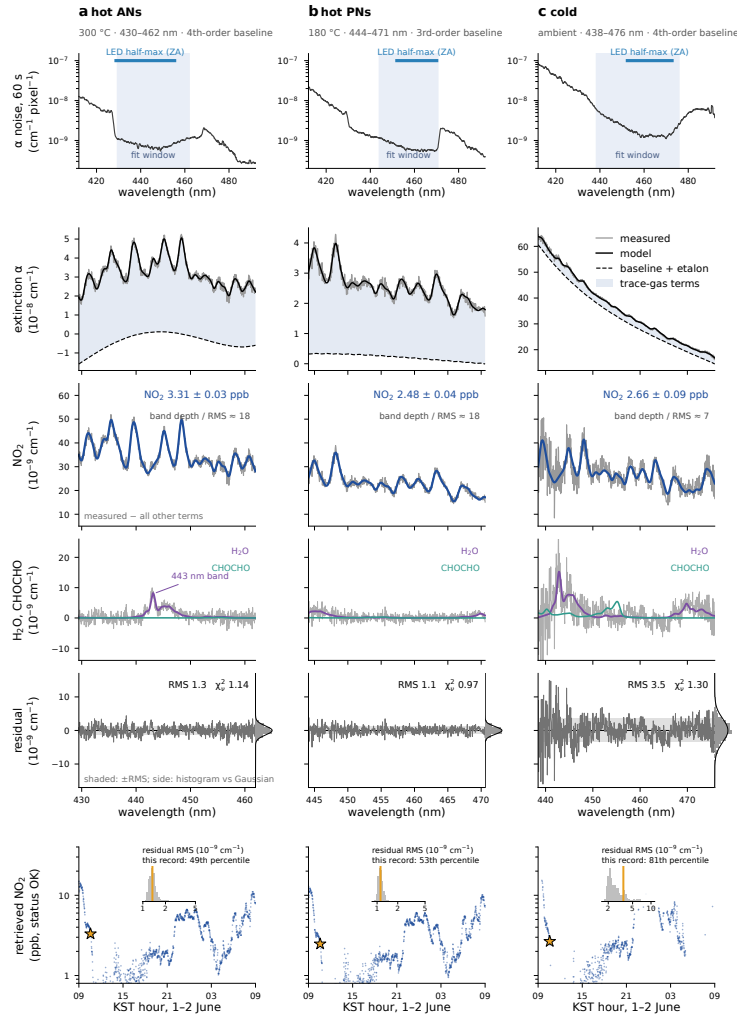


Figure 2. One record (1 June 2026, 10:35 KST; cold channel 10:34:34) retrieved in the three configurations of Table 1. Columns: **(a)** hot ANs (300 °C), **(b)** hot PN (180 °C), **(c)** cold. **Top row:** per-pixel noise of the 60 s extinction spectrum in the hour around the record, with the fit window shaded. Each window sits on the low-noise band set by its LED (bar: LED emission half-maximum from zero-air spectra); the cold window extends 13 nm below its LED half-maximum, where the noise is several times higher; the step at 428 nm in (a) is the band-pass filter edge and the step at 470 nm in (b) is the red edge of the 469 nm LED. **Middle block,** top to bottom: measured extinction with the fitted model and the baseline-plus-etalon term (shading: sum of trace-gas terms); the NO₂ contribution, shown as the measurement minus every other fitted term, with the fitted NO₂ term; the same for H₂O, with the fitted CHOCHO term; and the residual, with $\pm$ RMS shaded and its histogram against a Gaussian of the same RMS at right. The lower three rows share one vertical scale across columns. Values in the NO₂ row are the operational retrieval and its reported 1σ ; “band depth / RMS” is the peak-to-peak of the fitted NO₂ term divided by the residual RMS. **Bottom row:** the NO₂ retrieved by the same configuration over the surrounding 24 h (fits with reduced $\chi^2 \leq 1.5$; $n = 1397, 1399, 707$), the star marking this record; inset: distribution of residual RMS over those fits, on which this record sits at the 49th, 53rd and 81st percentile.

Table 2. Difference between the analytic Golub–Pereyra Jacobian and a central finite difference of the projected residual, evaluated inside a real fit, against the finite-difference step h , for the shift (px) and the squeeze. Bold: the minimum of each column.

step h	shift (px)	squeeze
10^{-1}	4.31×10^{-3}	outside the box
10^{-2}	4.29×10^{-5}	1.02
10^{-3}	4.29×10^{-7}	2.63×10^{-2}
10^{-4}	4.51×10^{-9}	2.61×10^{-4}
10^{-5}	2.60×10^{-9}	2.61×10^{-6}
10^{-6}	7.28×10^{-8}	2.61×10^{-8}
10^{-7}	5.79×10^{-7}	2.64×10^{-9}
10^{-8}	6.95×10^{-6}	2.87×10^{-8}
10^{-9}	1.09×10^{-5}	2.70×10^{-7}

pixel values of ± 0.37 px) crossed with squeezes of 0.998, 1.000 and 1.002 – the largest error in the retrieved NO_2 amount is 9.5×10^{-13} relative and the largest error in the recovered shift 6.7×10^{-10} px. The retrieval reproduces its own forward model to machine precision, which is a statement about the solver and not about the instrument.

250 The baseline order is the one structural choice the retrieval makes on its own. Sweeping it from 0 to 8 on a noise-free spectrum built with a fourth-order baseline, the retrieved NO_2 is in error by 27 % at order 0 and 0.75 % at order 1, and reaches machine precision at order 2 and above; the normalised condition number rises from 14.9 to 33.8 across the sweep and the degrees of freedom fall from 543 to 535. On noisy realisations of the same spectrum the scatter of the retrieved NO_2 is 0.88 to 0.91 times the reported 1σ at order 2 and above. For this configuration, with white photon noise and a correct model, the
255 reported uncertainty is accurate to about ten per cent – which is the baseline against which the structural terms of Sect. 4 should be read (Fig. 3a, b). It is a statement about one configuration: across the 13 synthetic instruments of Fig. 1a the white-noise ratio ranges from 0.91 to 1.32, and at nominal noise it lies between 1.22 and 1.32 in 9 of them, where the free wavelength shift adds scatter the covariance does not carry (Sect. 3.4).

3.2 The analytic Jacobian

260 The two-term Golub–Pereyra derivative of Sect. 2.3 is compared against a central finite difference of the projected residual, evaluated inside a real fit rather than on a constructed test function: the optimiser call is intercepted and the analytic Jacobian it was handed is differenced against the residual it was minimising. An invariant check of this form runs on every fit in the test suite at the solver’s default step; what is reported here is the step-size sweep (Table 2), which is what turns a passing check into a measurement.

265 Both parameters give the expected V (Fig. 3c). The truncation branch falls with a fitted slope of $+2.001$ in each, which is the h^2 of a central difference, and the round-off branch rises again as the difference of two nearly equal numbers loses significance.

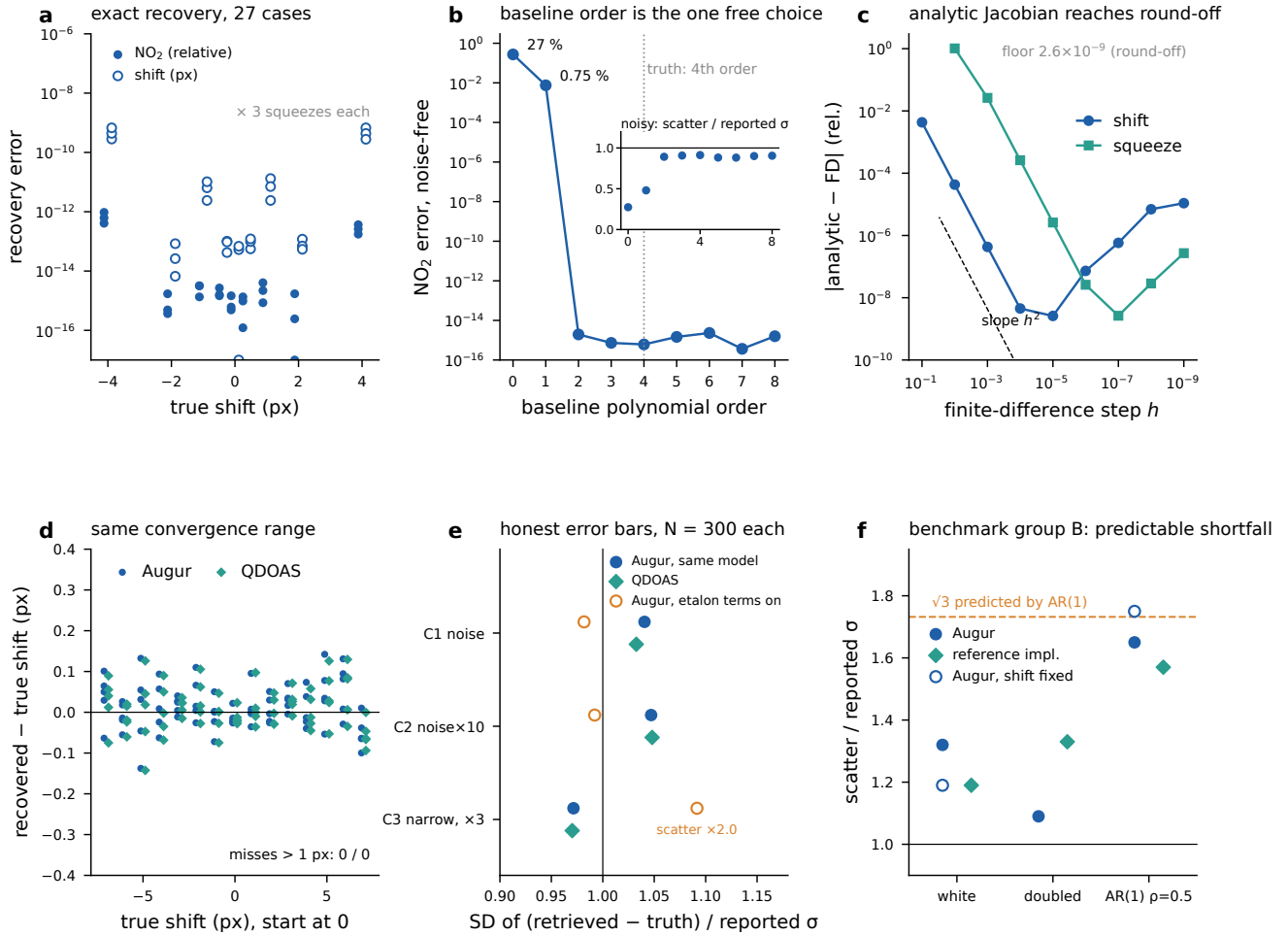


Figure 3. Validation of the retrieval core. **(a)** Error in the recovered NO₂ amount (filled, relative) and wavelength shift (open, px) for 27 noise-free spectra: nine shifts between -4 and $+4$ px, each with squeezes 0.998, 1.000 and 1.002. **(b)** Error in NO₂ against the order of the fitted baseline for a noise-free spectrum built with a fourth-order baseline; inset, on noisy realisations of the same spectrum, the realised scatter of NO₂ divided by the reported 1σ . **(c)** Difference between the analytic Golub–Pereyra Jacobian and a central finite difference of the projected residual, against the finite-difference step, for the shift and the squeeze; the dashed line has slope h^2 . **(d)** Shift recovered by Augur and by QDOAS minus the true shift, for true shifts from -7 to $+7$ px (five noisy realisations each, both started at 0). **(e)** Standard deviation of (retrieved $-$ true)/reported σ over 300 realisations in each of three conditions (measurement noise; ten times that; a narrow window with three times the noise), for Augur and QDOAS fitting the same model, and for Augur with its etalon terms switched on. **(f)** Benchmark group B: realised scatter divided by the reported uncertainty under white, doubled and autocorrelated (AR(1), $\rho = 0.5$) noise, for Augur, for the benchmark’s reference implementation and for Augur with the shift fixed at its true value; the dashed line is the $\sqrt{3}$ inflation predicted for $\rho = 0.5$.

The minima are 2.60×10^{-9} for the shift and 2.64×10^{-9} for the squeeze, a ratio of 1.018: the two parameters reach the same floor, which is the signature of a floor set by round-off in the shared residual evaluation rather than by any error in either derivative.

270 This is a stronger statement than agreement at one step size. An analytic expression that omitted a term – the one-term approximation of Kaufman (1975), which drops the derivative of the projector, is the natural candidate – would agree with the finite difference only down to the size of the omitted term and would then flatten, not descend to the round-off floor. The descent is what identifies the expression being differentiated as the derivative itself.

The two parameters must be swept separately. The shift is measured in pixels and is $O(1)$, while the squeeze is a multiplier
 275 near unity constrained to a box of order ± 0.02 , so $h = 10^{-2}$ is already half the box for the squeeze and the entry there is a bound artefact rather than a failure of the derivative. A combined maximum over both parameters would show 1.02 at that step and hide the V.

3.3 An independent implementation of the same problem

QDOAS (Danckaert et al., 2017) solves the same separable least-squares problem by a different route: Marquardt–Levenberg
 280 with a singular-value decomposition and a numerically differenced Jacobian, against Augur’s variable projection with an analytic projected Jacobian. It does not know about cavities, so the comparison is made by presenting the extinction spectrum as a virtual transmission, $I = \exp(-K\alpha)$ with $I_0 = 1$, whereby the optical depth QDOAS forms is exactly $K\alpha$ and the two codes are solving the same equations on the same numbers.

On noise-free synthetic spectra with shifts of 0, -1.96 , $+2$ and -4 px, Augur recovers NO_2 to 9.5×10^{-13} relative and
 285 QDOAS to 5.0×10^{-3} , the latter limited by its output resolution rather than by its algorithm; QDOAS recovers the shift to within 0.0042 px. On 30 noisy realisations the two codes have root-mean-square relative errors of 0.0042 and 0.0045 against the same truth, and differ from each other by at most 5.7×10^{-3} relative in NO_2 . On the field record (53 days), with each spectrum’s measured temperature and pressure used for the unit conversion and records selected by an NO_2 -conditional χ^2 envelope, the NO_2 amounts of the two codes agree with $r^2 \geq 0.990$ in all three configurations (Table 3). The agreement is not
 290 a scale identity: the Deming slope QDOAS/Augur is 1.039 in the cold configuration and 1.063 and 1.043 in the 180 and 300 °C configurations, where $r^2 = 1.000$, so the two codes differ in the heated-channel NO_2 scale by 4–6 % for a reason we have not identified. The weak absorbers agree less well (CHOCHO r^2 0.889–0.959, slope 0.81–1.19; H_2O slope 1.04–1.38).

In a larger test with 300 realisations in each of three noise conditions, the standard deviation of the error normalised by the reported uncertainty is 0.97 to 1.05 for both codes when they fit the same model, and started from zero both recover true shifts
 295 between -7 and $+7$ px with no miss larger than a pixel (Fig. 3d, e). Switching on Augur’s etalon terms in the narrow window doubles the scatter there – a choice of model, not a property of the solver.

The conclusion drawn from this in Sect. 2.9 is therefore narrower than identity: on synthetic spectra the two codes return the same concentrations, and on field spectra they agree in NO_2 to within a scale difference of a few per cent. Where they diverge most – the wavelength shift in the cold window, and with it the weak absorbers there (Sect. 5.1) – the data do not determine
 300 the shift, so neither code’s value is established by its fit.

Table 3. Agreement of QDOAS with Augur on the field record, per retrieval configuration and species: number of records n in the NO_2 -conditional χ^2 envelope, Deming slope QDOAS/Augur ($\lambda = 1$) and r^2 . The unit conversion uses each spectrum’s measured temperature and pressure.

configuration	species	n	Deming slope	r^2
cold	NO_2	36 241	1.039	0.990
cold	CHOCHO	36 241	0.807	0.889
cold	H_2O	36 241	1.377	0.887
hot PNs (180 °C)	NO_2	73 916	1.063	1.000
hot PNs (180 °C)	CHOCHO	73 916	1.195	0.914
hot PNs (180 °C)	H_2O	73 916	1.042	0.974
hot ANs (300 °C)	NO_2	74 539	1.043	1.000
hot ANs (300 °C)	CHOCHO	74 539	1.054	0.959
hot ANs (300 °C)	H_2O	74 539	1.038	0.981

3.4 A benchmark that others can run

The comparison of Sect. 3.3 is specific to one pair of codes and one set of cross sections. To make the same measurements repeatable elsewhere we release a synthetic benchmark of 261 cases in six groups: exact recovery with no noise (A), reported uncertainty against realised scatter under white, doubled and autocorrelated noise (B), bias from a deliberately incomplete
305 model (C), cases whose wavelength shift is not determined by the data (D), the same truth presented in two fitting windows (E), and the cost of the sub-pixel interpolator against band-limited truth (F). Truth is known by construction, the spectra are built from the instrument’s own cross sections and wavelength calibration but contain no retrieval code, and the generator is deterministic under a stated seed.

Group B is worth reporting in full, because its result is not the one the paper’s own closure test gives and the difference is
310 instructive. Augur returns scatter-to-reported ratios of 1.32, 1.09 and 1.65 for white, doubled and autocorrelated noise, against 1.19, 1.33 and 1.57 from the reference implementation: both codes pass one condition of three, and they fail different ones. Two effects account for the pattern. Fixing the wavelength shift at its true value brings Augur’s white-noise ratio from 1.32 to 1.19, which is the reference implementation’s value, so the difference between the two codes under white noise is the nonlinear degree of freedom and the size of the box it is searched in, not the covariance they propagate. And the correlated-noise result is
315 predicted by the noise model rather than being a property of either code: an AR(1) residual with $\rho = 0.5$ inflates the variance of a sum by $(1 + \rho)/(1 - \rho) = 3$, so a covariance built on an independence assumption must understate the scatter by $\sqrt{3} = 1.73$. The measured 1.57 and 1.65 sit just below it, and fixing the shift moves Augur to 1.75 – the free shift had been absorbing part of the correlated structure (Fig. 3f).

This is the benchmark doing what it was built for. The shortfall under correlated noise is not a defect to be repaired in either
320 implementation; it is the reported uncertainty answering a question about photon statistics when the residual is not photon
statistics, and its size is calculable in advance.

A reference implementation is included, and its scorecard is the example against which a submission is read. It is a deliberately plain profiled grid search, not a production retrieval, and the benchmark is not a ranking: groups C and E are reported rather than passed or failed, because the quantity they measure is not one any implementation is expected to cover.

325 3.5 Relative scale between the two hot channels

The alkyl nitrate product is a difference between two channels, so their relative scale enters it directly. NO₂ was injected into both heated channels, with both ovens at their operating temperatures, at two concentration steps on 2026-08-10/11. The steps did not equilibrate (the delivery line was still passivating), so only the ratio between the channels is used. In the original processing the median of the per-record ratios was 1.2278 at the lower step and 1.2276 at the higher (the median channel
330 readings were 14.75 and 11.97 ppb, and 71.57 and 58.16 ppb; ratios of these medians are 1.2322 and 1.2306), and an origin regression over the 58 one-minute means of one file gives a slope of 0.81666 with r^2 0.9998. Reprocessed with the current code, block 2053 reads 1.239 times block 4101 (interquartile range 1.231–1.255, $n = 41$ records with block 4101 above 20 ppb).

This establishes that the relative scale is constant over the range tested. It does not give the scale used in the campaign product: the laboratory ratio describes the calibration state of August, and the product uses $g' = 0.952$ for 18 May–5 June,
335 from a field injection. Nor does it establish the absolute recovery of either channel, or separate optical from thermal causes of the offset between the channels, because both cells were at their operating temperatures throughout.

3.6 What is not verified here

A comparison with a co-located reference analyser exists, and it bounds what this paper can claim. Against the NIER air-quality-monitoring NO₂ analyser at the site (molybdenum-converter type), the operational NO₂ of the heated channels has
340 ordinary-least-squares gains of 0.57 (180 °C) and 0.65 (300 °C), while the ambient-temperature channel has 0.92–1.00; the median ratio of heated to ambient-temperature NO₂ is 0.72, although the heated channels measure NO₂ plus thermally produced NO₂ and should read at least as high. A laboratory injection with the ovens at 300 and 180 °C gave a block-2053 sensitivity of at least 1.08 relative to the cylinder, so oven temperature alone does not explain the discrepancy, and its cause is unresolved. It is a multiplicative error that leaves the residual, the reported uncertainty and every context field proposed in
345 Sect. 6 unchanged – an instance of the class of error this paper is about, not a property of the spectral fit. No absolute accuracy is claimed. The verification of Sects. 3.1 and 3.3 is of the retrieval against its own forward model and against another implementation of that model, which bounds implementation error but not model error – and model error is precisely what Sect. 3.4 measures on the benchmark.

4 Structural error budget

350 4.1 What the fit reports, and what it cannot report

The covariance returned by the fit (Sect. 2.5) propagates one thing: the residual scatter of the spectrum being fitted. For the operational configuration this gives a single-scan reported uncertainty of 0.55 % on the channel NO₂ amount, and it is the number that appears in the data product.

That number is conditional on the forward model being correct. Three inputs of Eq. (1) are not measured simultaneously with
355 the spectrum, but interpolated to its time from neighbouring calibration blocks: the zero-air reference I_0 , the mirror reflectivity $R(\lambda, t)$, and – for the periodic optical structure – the etalon frequency f , which is held at a fixed value rather than redetermined per scan. An error in any of these enters α directly and is therefore indistinguishable, within a single fit, from a change in the gas amounts. The residual does not see it, so the reported uncertainty does not contain it.

Two further inputs were tested. Sample temperature and pressure, interpolated the same way as I_0 , contribute below 0.1 %.
360 For the instrument line shape the test is weaker than it looks. The kernel is resampled onto the detector grid with a radius of $\lfloor 4\sigma \rfloor + 1$ pixels, and at the mismatch measured in the ambient-temperature channel (FWHM 0.765 against 0.785 nm) the retrieved amount moved by $\sim 10^{-5}$ %; this shows that the test did not carry a width change of that size into the forward model, not that the retrieval is insensitive to it. A test with a continuously resampled kernel has not been made, and no mismatch was measured for the heated channels. The residual does carry a signal-proportional model error: the median reduced χ^2 of the
365 300 °C channel rises from 1.05 at NO₂(180 °C) below 1 ppb to 1.57 at 10–20 ppb. That error is not in the budget below; the cross sections and the line shape are candidates.

4.2 Measuring a structural term

A structural term is measured by perturbing the input the way the operational pipeline can actually be wrong, then re-running the full retrieval and reading the change in the retrieved amount. Two perturbations are used.

370 **Leave-one-out (LOO).** One calibration knot is removed, the remaining knots are re-interpolated across the resulting gap, and every scan in that gap is re-retrieved. This is an upper bound by construction: the interval the interpolant must span is roughly doubled relative to operation.

Split-half. Each calibration block is split into two halves – contiguous (first and second half) or interleaved (even and odd scans) – and two independent block means are formed. Their difference measures the noise on a knot with no change of interval,
375 and therefore requires no extrapolation back to the operational spacing. With $n/2$ scans per half the difference carries a factor of two relative to the full-block knot noise, which is divided out.

The two disagree by a channel-dependent factor, and that disagreement is itself the result (Table 4).

A ratio near unity means the LOO error is entirely knot noise – the interpolant is not being asked to bridge any real curvature, and halving the reference interval would not improve it. A ratio near one half means the interval does carry curvature. The
380 ordering agrees with the direct measurement of Sect. 6.2: tripling the reflectivity interval from 2 to 6 h raises the interpolation error by 15 % in hot PNs and by 33 % in hot ANs.

Table 4. Zero-air reference-interpolation error in the channel NO₂ amount (per cent) from the split-half estimate of knot noise, with contiguous and interleaved halves, and from leave-one-out (LOO) knot removal, and the ratio split-half/LOO. Hot channels: the 157 zero-air knots of 18–24 May, current fit settings and clock-aligned reflectivity. Cold: an earlier run with different settings, not recomputed on this basis.

channel	split-half, contiguous	split-half, interleaved	LOO	ratio
hot PNs	1.437 %	1.782 %	1.95 %	0.74–0.91
hot ANs	0.90 %	0.97 %	2.25 %	0.40–0.43
cold (earlier run)	–	2.391 %	3.95 %	0.61

This bears on a recommendation made qualitatively by Washenfelder et al. (2008). For the 180 °C channel the interpolation error is mostly knot noise (ratio 0.74–0.91): measuring the references *more often* would reduce it little, and averaging more scans per calibration block is the lever that could. For the 300 °C channel (0.40–0.43) the interval carries curvature, and shorter
385 intervals would help. Contiguous and interleaved halves give similar knot noise in the NO₂ amount (the contiguous estimate is 7–19 % smaller), so the interleaved split does not understate it.

4.3 Terms must be propagated jointly, and the scale taken once

It is standard to measure each term separately and add the results in quadrature. Here that is wrong twice over, and the two failures have different causes.

390 **Model-space and α -space perturbations are not additive.** Perturbing I_0 and perturbing R both move α , and their effects add scan by scan: the median of $|\delta_{\text{joint}} - (\delta_{I_0} + \delta_R)|$ is 0.8–2.2 % of the signal. The etalon is different. Its sine and cosine columns are free linear parameters, so they absorb part of any smooth perturbation of α before it can reach the gas columns. Perturbing all three simultaneously gives a smaller result than summing the three responses, by 8.3 % (hot ANs), 6.4 % (hot PNs) and –0.2 % (cold). Adding the etalon term as if it were independent therefore overestimates, and by an amount that
395 depends on the fitting window and polynomial order.

Neither variances nor robust scales add. Summarising each term by a median-based scale and combining those in quadrature misses the directly measured joint scale by +13.3 % (hot ANs), –8.1 % (hot PNs) and +6.9 % (Σ ANs). Performing the identical combination on standard deviations does not repair it: the joint exceeds the quadrature sum by 15.5 % for the ANs channel and 7.6 % for Σ ANs, and falls 2.5 % below it for PNs. Neither space is additive, and neither discrepancy has a
400 predictable sign.

For variances the failure has two measured parts. The single-term responses of Σ ANs are correlated (pairwise $r = 0.25$, 0.18 and 0.03), which adds 28.7 % to the sum of their variances, and the terms do not superpose – the etalon interaction above – which removes 12.9 %, leaving the joint variance 15.8 % above the sum. Robust scales fail in addition through distribution shape. Write s for the normalised shape statistic 1.4826MAD/SD, which is unity for a Gaussian and 0.727 for a Laplace. The
405 three Σ ANs components have $s = 0.73$ (I_0), 0.32 (R) and 0.98 (etalon): one is Gaussian, one Laplace-like, one very much heavier-tailed. A median-based scale therefore captures a different fraction of the variance in each, and the joint series has a

Table 5. Structural error budget, ppb, from a single propagation over the 60 s ambient records of 2026-05-18..05-24 on the clock-aligned reflectivity reference (Sect. 5.4): of the 9034 records with a reflectivity knot within 1 h, the 187 in which a fit of either channel terminated at a bound are excluded (Sect. 5.2), leaving 8847. Σ ANs is formed as in the product, $300\text{ }^{\circ}\text{C}$ minus g' times $180\text{ }^{\circ}\text{C}$ with $g' = 0.9524$ for this period. Columns I_0 , R , f are single-term responses, shown for diagnosis; *joint* is the measured simultaneous response and is the quantity to be used. Both a robust scale (1.4826MAD) and a standard deviation are given: the first describes a typical record, the second includes the tail. s is the shape statistic of Sect. 4.3.

		I_0	R	f	joint
hot ANs channel	rSD	0.0437	0.0350	0.0082	0.0641
	SD	0.0600	0.0829	0.0086	0.1186
	s	0.73	0.42	0.95	0.54
hot PNs channel	rSD	0.0416	0.0126	0.0187	0.0435
	SD	0.0428	0.0384	0.0189	0.0590
	s	0.97	0.33	0.99	0.74
Σ ANs	rSD	0.0558	0.0269	0.0191	0.0693
	SD	0.0768	0.0831	0.0195	0.1236
	s	0.73	0.32	0.98	0.56

fourth shape again ($s = 0.56$). A variance is indifferent to shape, so for variances the discrepancy is covariance and interaction alone.

What makes this fatal for quadrature is not the size of the error but its unpredictability. The discrepancy is +13 % in one channel and −8 % in another, and moving to variances neither removes it nor changes its sign − +15.5 %, −2.5 % and +7.6 % for the same three series – so it cannot be absorbed into a correction factor: knowing the factor requires knowing the covariance, interaction and shape of the joint response, and that is available only after the joint propagation has been performed. Quadrature of robust scales cannot be repaired, only replaced. The rule used throughout this paper is therefore:

Propagate every term, simultaneously, into the final product time series. Take the scale once, at the end, with the same estimator used for the quantity it will be compared against.

4.4 The budget

Table 5 gives the budget as the robust scale of the jointly propagated response, in mixing-ratio units. The three hot/cold entries are per channel; the Σ ANs row is propagated through the channel difference, not formed from the channel rows, for the reason given in Sect. 4.3.

Which term dominates depends on which scale is read, and the two answers are opposite. Of the single-term quadrature sum for Σ ANs, I_0 carries 74 % on the robust scale but 45 % in standard deviation, while R carries 17 % and 52 %. The reflectivity term is small for a typical record and the largest contributor overall, because its error is concentrated in a few intervals: five

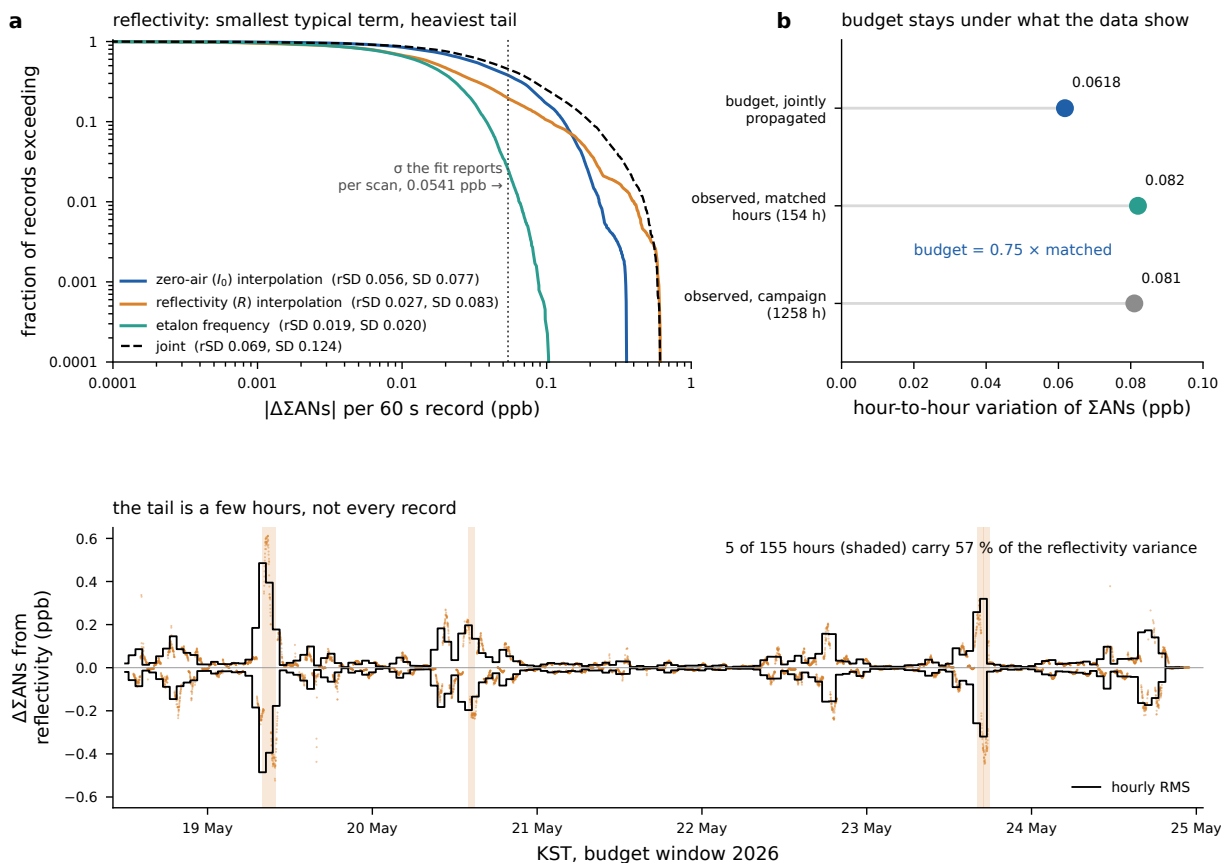


Figure 4. The structural error budget for the alkyl nitrate product, from leave-one-out propagation over the 8847 records of the budget window (18–24 May). **(a)** Fraction of records whose ΣANs changes by more than a given amount when each reference term is perturbed alone, and when all three are perturbed together; the legend gives the robust scale and standard deviation of each (Table 5), and the dotted line is the uncertainty the fit reports for one scan. **(b)** Hour-to-hour variation of ΣANs predicted by the jointly propagated budget, against the variation observed in the matched hours and over the whole campaign. **(c)** The reflectivity term record by record (points) and as an hourly RMS (steps); shaded, the five hours that carry 57 % of its variance. ΣANs responses are formed with $g' = 0.9524$, as in Table 5.

of the 155 hours in the window carry 57 % of its variance (Fig. 4a, c). Section 6 takes this as the basis for choosing between remedies.

425 Records whose perturbed fit terminated at a parameter bound are excluded, for consistency with Sect. 5.2: a fit pinned at a bound has not converged, and its response is not a measurement of anything.

The cold channel is not included in Table 5. It was not re-run on the all-record basis, which lies outside the critical path of this paper, and its earlier representative-record values are withdrawn rather than reported on a different basis.

Table 6. Candidate denominators for the ratio of total to reported Σ ANs uncertainty: value (ppb), time resolution, and the ratio of the total uncertainty (Table 5, robust scale to standard deviation) to each. Bold: the denominator used. The zero-air value is approximate (Sect. 7.2).

denominator	ppb	resolution	total / denominator
fit-propagated σ, per scan	0.0541	60 s = 1 scan	1.63–2.50
uncertainty column shipped with the product	0.0806	5 min, systematic	1.32–1.83
fit-propagated σ , 5 min bin	0.025	5 min	not comparable
zero-air scatter, SD	$\approx$ 0.086	34-scan block means	not comparable

The record evaluated here is the product reprocessed on the clock-aligned reflectivity reference (Sect. 5.4). It is not the
430 dataset delivered to the campaign, which was produced before that correction and is retained as delivered.

Table 5 is conditional on a population. A perturbation experiment can only be run on scans that have neighbouring knots of both reference types, so periods in which the calibration itself failed are absent from it – and those are precisely the periods with the largest errors. How far each record sits from its nearest knot of each type, how often the interpolation is in fact an extrapolation, and which periods those are, is measured campaign-wide in Sect. 5.3. The budget window of this section is the
435 best-covered stretch of the deployment: no record outside the knot range and no gap longer than two hours, against 11.03 % of cold records in gaps longer than two hours over the campaign as a whole. Table 5 therefore describes the instrument when its calibration is working, and Sect. 5.3 measures how often that condition does not hold.

Against what the fit reports the budget is large. A budget record is one 60 s ambient bin, which at the operational scan rate is one scan, and the median fit-propagated Σ ANs uncertainty at that resolution, over the same records, is 0.0541 ppb.
440 The structural contribution alone is therefore 1.28 times the reported sigma on the robust scale and 2.29 times it in standard deviation, and the total uncertainty is **1.63 to 2.50 times what the fit reports**, the range spanning the two scales rather than any doubt about the measurement. Because leave-one-out doubles the interpolation interval (Sect. 4.2) and the window is the best-calibrated week of the deployment, these are upper-bound estimates for a well-calibrated period.

The denominator has to be argued for, because three other quantities are available and none of them answers this question
445 (Table 6).

The column shipped with the product is not a measurement uncertainty. It is a gain systematic: regressed against the channel NO_2 amount over 14 974 records it is $0.04209 [\text{NO}_2] + 0.00025$ ppb with $r = +0.9997$, and the product header states that it covers day-to-day inter-channel gain only and must not be used alone to assess detection significance; it does not bound the absolute scale either (Sect. 3.6). Comparing the structural budget against it is a defensible sentence, but a different one: it asks
450 whether structure is comparable to the gain term, not whether it exceeds the precision the retrieval claims. The 5 min row is the fit uncertainty averaged over five scans – over the deployment it is $0.0558/\sqrt{5}$ – and dividing it into a structural scale measured at 60 s would mix resolutions in the opposite direction. The zero-air scatter is the honest precision floor, but it is measured on 34-scan block means, each retrieved against a reference interpolated across the removed block, so it combines a different scan count with reference-interpolation error and is not a single-scan quantity. The per-scan fit uncertainty is the median, over 8832

Table 7. Hour-to-hour variation of hourly Σ ANs, summarised by $1.4826 \text{ median}|\Delta|/\sqrt{2}$ (ppb), observed and predicted by the budget, and the ratio of the budget to the observed statistic in the matched hours. Observed: the clock-aligned Σ ANs product (quality flag below 2), hourly means of at least 8 records.

	gate statistic (ppb)	ratio
observed, campaign (1258 h)	0.081	–
observed, hours matched to the budget (154 h)	0.082	–
budget, hourly, jointly propagated, matched hours	0.0618	0.75
budget, at record resolution	0.0693	0.84

of the budget records, of the NO₂ uncertainty written by the operational fit of each channel propagated through the channel difference, $\sqrt{\sigma_{300}^2 + g'^2\sigma_{180}^2}$. Over the whole deployment, with the g' of each calibration state, the same median is 0.0558 ppb (total 1.59–2.43 times), and at five-minute resolution the fit uncertainty lies between 0.0217 and 0.0326 ppb (5th and 95th percentiles).

4.5 Consistency with the observed variation

An error budget assembled from perturbation experiments can be checked against the data it claims to describe. If a term is channel-differential, it must appear in the channel-difference product; if it varies on the timescale of the calibration interval, it must appear in the hour-to-hour variation of that product. The observed variation is an upper bound on the measurement error, because it also contains real atmospheric variability.

The comparison is only meaningful if both sides pass through the same estimator. The observed quantity is summarised by $1.4826 \text{ median}|\Delta|/\sqrt{2}$ over adjacent hourly means, so the propagated series is summarised the same way.

The budget series is itself hourly (median spacing between consecutive entries 1.00 h, p_{90} 1.03 h), so the two series describe the same lag, and the comparison is made on the same timestamps, paired.

Two comparisons are defensible, and they differ because the perturbation series and the observed series have opposite temporal correlation. Their measured lag-1 autocorrelations are -0.60 for the budget and $+0.73$ for the observation in the matched hours. The negative value is a property of the experiment, not of the instrument: removing a knot pins the interpolant at the two neighbours and forces it to depart on alternate sides, so the perturbation response alternates in sign from hour to hour. The positive value is atmospheric persistence.

The comparison is made at the resolution at which the observed bound is defined – hourly means, differenced – and against the same hours (Table 7).

The last row uses a different operator and is given only to show the size of that choice; both ratios in the column are taken against the same matched denominator, 0.082 ppb. Both the numerator and the denominator of the ratio quoted here are evaluated on the same 154 matched hours and 153 consecutive pairs; the campaign-wide denominator gives almost the same ratio, 0.76.

The budget predicts **0.75 times** the observed hour-to-hour variation (Fig. 4b). It does not exceed it, as it must not, and it is not far below it either, which would have implied that the observed variation is dominated by something the budget does not describe.

We report the two as consistent, and assign no pass or fail. The test cannot do better: it assumes the error is white at one-hour lag (any slower structural component is invisible to it), it cannot separate measurement error from atmospheric variability, and the upper bound doubles the interpolation interval by construction. The choice of operator alone moves the ratio from 0.75 to 0.84 (Table 7); differences of that size are the resolution of the method.

One correction must be avoided, because it is available and wrong. Since $\text{Var}(\Delta) = 2\sigma^2(1 - \rho)$ is an identity when ρ is the lag-1 autocorrelation of the same series, “correcting” the observed increment statistic for ρ returns the standard deviation of the observed series itself – 0.366 ppb for the hourly Σ ANs, i.e. the full geophysical variability. Comparing a structural budget against that number is not a test; the differencing was introduced precisely to remove it.

4.6 What should be reported instead

For each retrieved amount the product should carry three numbers, not one: the fit-reported uncertainty; the structural contribution evaluated for that scan’s calibration context; and a flag identifying scans whose calibration knots are themselves suspect. The third matters because the structural error is not uniform – the R term carries the smallest shape statistic in Table 5 ($s = 0.32$ for Σ ANs), meaning a small number of intervals carry most of its variance. Section 6 specifies all three and states which are implemented. Sections 5.3 and 5.4 add two further fields of the same kind – how far the record is from its calibration, and whether the reference it used was still valid – which the pipeline can supply without any new measurement. A further step, not attempted here, would treat the zero-air reference and the reflectivity as measured quantities with their own covariance and propagate the structural term inside the fit rather than afterwards.

5 What the pipeline already knows and does not report

Section 4 measured what the fit cannot know: errors that enter α through inputs the fit never sees, and which therefore leave no trace in the residual it minimises. This section is about the other category, and it is the cheaper one to fix. In each case below, the information needed to qualify a retrieved amount already exists somewhere in the pipeline – in the profiled objective, in the optimiser’s termination state, in the knot table, in the timestamps of the files being read – and is discarded before the product is written. Nothing here calls for a better retrieval algorithm or an additional measurement. It calls for the pipeline to report what it has already computed.

5.1 A nonlinear parameter the data do not determine

A nonlinear least-squares routine reports convergence when its step and tolerance criteria are met. That is a statement about the optimiser, not about the objective: an iteration that stops because the step became small has established that it is at a stationary point of the profiled residual, and nothing more. If the profiled residual is nearly flat over the whole range the parameter is

510 allowed to take, the returned value is set by where the search started and where the bounds are, and the uncertainty computed from the curvature at that point describes a curvature that is not there.

This is not a defect of any particular implementation. It follows from reporting a parameter value without reporting how much the data constrain it, which is standard practice in DOAS retrieval software.

A measured case: the wavelength shift in the cold channel. The wavelength shift is the natural test because it is the
515 parameter the retrieval is least able to distinguish from a change in the background. Scanning the shift across ± 4 px and profiling out the linear parameters at each point gives the objective's actual dependence on it.

In the cold channel the objective is flat. For twelve ambient spectra scanned across ± 4 px, the ratio of the largest to the smallest profiled residual sum of squares has a median of 1.0120 and never exceeds 1.0611; in a larger set of 165 spectra the minimum lies at an edge of the scanned range in 77 % of cases, and evaluating the objective at zero shift rather than at its
520 minimum costs between 0.01 % and 4.35 %. The same scan in the hot PNs channel spans a factor of 3.09×10^5 .

Two implementations confirm that the consequence is not numerical. Under matched settings the shift returned for the same spectra is -2.014 px in this work, with 85.5 % of fits terminating at a bound, against -0.088 px from QDOAS with 4.5 % at a bound; tightening the QDOAS convergence threshold to 10^{-12} changes nothing. Both codes converge, both report a value, and the values differ by two pixels because the data do not distinguish them.

525 The same channel does distinguish them when the absorber is strong enough. On 2 June NO_2 from a cylinder standard (nominally $10 \mu\text{mol mol}^{-1}$ in N_2 , past its certification date and used here only for its spectrum) was added to zero air in the cold cavity for half an hour; retrieved against the zero-air spectrum that preceded it, with the operational reference and window, the plateaus at 9.3 and 3.6 ppb place the minimum of the objective at -0.10 and -0.15 px, and evaluating it at -2.0 px instead costs 39 % and 7.5 % of the residual, against nothing at all for the zero-air spectrum itself. The QDOAS value lies within
530 0.06 px of that minimum; the value returned here does not, and the operational fixed value of -0.5 px costs 2.0 % at 9.3 ppb. What the ambient spectra lack is not a defined shift but the column to locate it (Fig. 5b, c).

What the retrieved amounts do with that freedom is the point. Comparing the cold-channel retrievals of this work with those of QDOAS on the same 5364 spectra, the regression slopes between the two codes are 0.940 (NO_2), 0.858 (CHOCHO) and 1.027 (H_2O) when Augur's shift floats, and 1.019, 0.662 and 1.178 when it is fixed at -0.5 px. The weakest absorber in the
535 window moves by 23 % according to a parameter the residual cannot resolve.

A code comparison cannot say which treatment is right. For two codes solving the same problem every species should give unity, and none does in either treatment; with the shift fixed, CHOCHO and water still differ between the codes by 34 and 18 %, so the disagreement is not the shift alone. Choosing a convention silently is what current practice does. The laboratory spectrum answers what the ambient ones cannot: the shift this channel supports is close to zero, -0.10 px. The operational fixed
540 value is 0.4 px from it, and replacing -0.5 by -0.1 px changes the slopes against QDOAS by at most 2.2 % (NO_2 -0.06 %, CHOCHO $+2.2$ %, H_2O -0.8 %); the floated value is 1.9 px from it and moves CHOCHO by 23 %. What divides the results is whether the shift is fixed, not the value it is fixed at, and neither fit report could have said so. What locates the parameter is a spectrum with enough absorber; what the product should carry is the value such a spectrum gives, and where it came from.

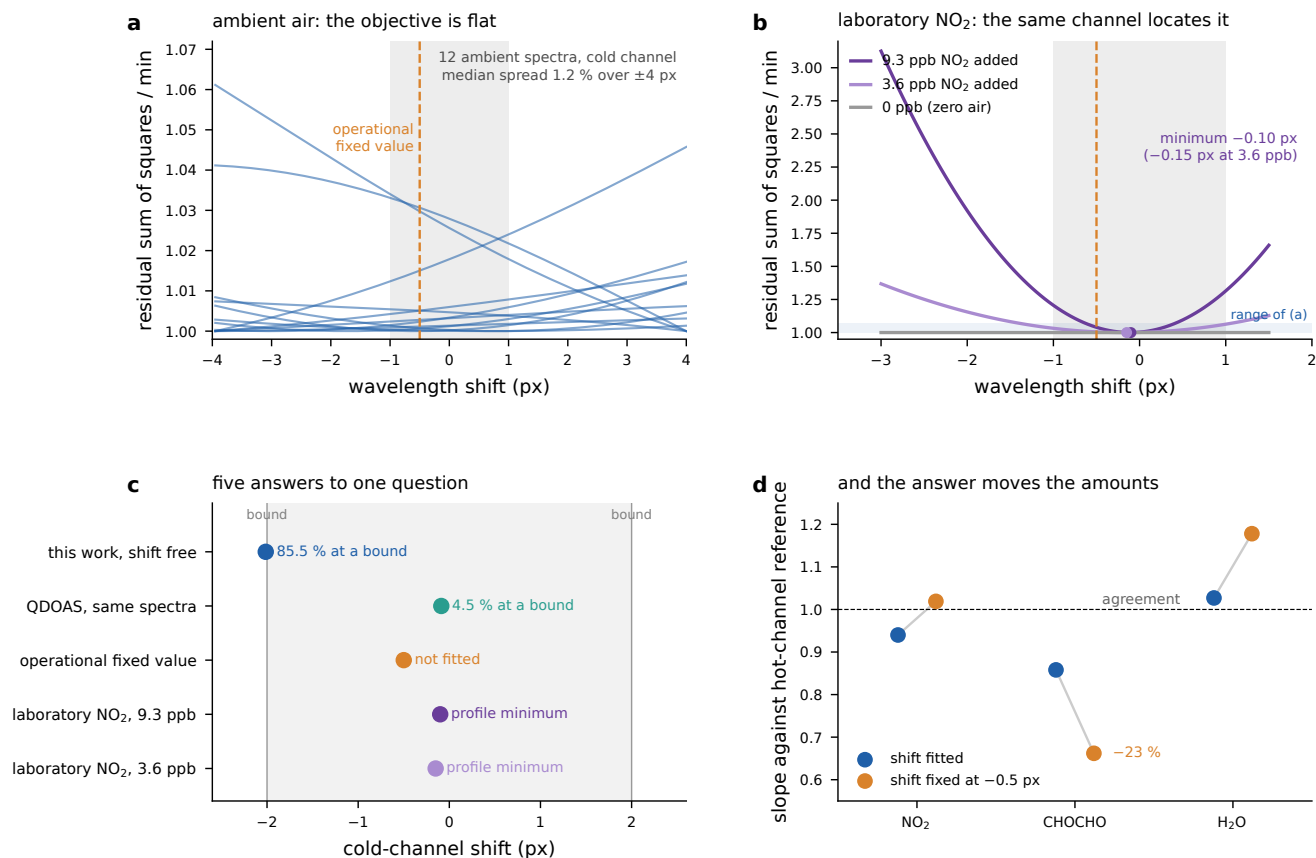


Figure 5. The wavelength shift in the cold channel is not determined by ambient spectra, and is determined by a laboratory spectrum with added NO₂. **(a)** Profiled residual sum of squares against shift for twelve ambient spectra (the edge-minimum fraction quoted in Sect. 5.1 is over 165), each normalised to its own minimum; shaded, the range the operational configuration allows; dashed, the value it fixes. **(b)** The same scan for the laboratory spectra of 2 June: zero air with 9.3 and 3.6 ppb of NO₂ added, and zero air alone; the band marks the vertical range of (a). **(c)** Shift returned for the ambient spectra by two independently converged implementations under matched settings (with the fraction of fits at a bound), the operational fixed value, and the profile minima of (b). **(d)** Slope of the regression between the cold-channel retrievals of this work and of QDOAS on the same spectra ($n = 5364$), with Augur’s shift fitted and fixed; for two codes solving the same problem every species should give unity.

What is degenerate with what. Not every input error threatens the same parameter. Propagating the three structural terms of Sect. 4 separately and recording how far each moves the wavelength shift separates them cleanly (Table 8).

An error in the reflectivity changes the magnitude of the extinction and essentially nothing else – it moves the shift by a median of 0.006 px (90th percentile 0.027 px), although the largest displacement reaches 0.285 px. An error in the zero-air reference, or in the etalon frequency, changes the shape of the extinction and is therefore partly absorbed by the axis. This is

Table 8. Change in the fitted wavelength shift (px) when each structural term of Sect. 4 is perturbed alone, and when all three are perturbed together: median, 90th percentile and maximum over the 8847 clock-aligned budget records of Table 5 (for each Σ ANs record, the larger of the two channels’ changes).

perturbation	median	p_{90}	max
zero-air reference	0.187 px	0.826	4.388
reflectivity	0.006	0.027	0.285
etalon frequency	0.154	0.347	2.941
all three, joint	0.243	0.766	4.276

the same distinction that Sect. 6.1.1 draws from the shape statistic, arrived at independently: the reflectivity is a pure amplitude
550 term, and the other two compete with the nonlinear parameters.

It also bounds the concern that the budget of Sect. 4 is really measuring basin hopping. Records whose shift moved by more than a pixel under perturbation are 6.2 % of the total and carry 5.0 % of the variance, and removing them raises the budget rather than lowering it (rSD 0.0693 to 0.0706 ppb, SD 0.1236 to 0.1244 ppb). The large-shift records are the low-amplification ones, where the absorption is deep and the retrieval is well conditioned.

555 **What the pipeline could report instead.** The profiled scan above is too expensive to run per spectrum, but its conclusion is available cheaply from the ratio of the objective at the returned value to the objective at a fixed reference point – a quantity the fit evaluates anyway. Where that ratio lies within the noise, the parameter should be fixed, said to be fixed, and the cost of fixing it carried in the uncertainty of every amount that depends on it. Reporting -2.014 ± 0.03 px for a quantity the data locate only to ± 2 px is the failure this subsection describes, and for this channel the resulting term is larger than any in Table 5. Because
560 the shift belongs to the spectrograph rather than to the air, it could instead be fitted jointly across the spectra of a calibration block – a separable problem with common nonlinear parameters; this is not implemented here, and it would need an absorber in the block, as in the laboratory spectrum of Fig. 5b.

5.2 Termination at a bound is a routine outcome

Bound termination is usually treated as a failure to be logged. In this instrument it is the normal state of one channel and a
565 systematic response to perturbation in another.

The operational configuration (Appendix A) allows the shift different freedom in each channel – $[-10, +0.5]$ px in hot ANs and $[-5, +5]$ px in hot PNs – and fixes it at -0.5 px in the cold channel. In a sensitivity run that instead bounded the cold shift to $[-1, +1]$ px, every one of the 22 base fits returned exactly -1.000 px, the bound itself, and was reported as converged. In the production propagation of Sect. 4 the cold shift is held fixed, as in operation, and moves by exactly zero in every record,
570 which confirms the configuration but also means the channel contributes no information about the parameter at all.

On the clock-aligned basis of Table 5, 187 of the 9034 Σ ANs records are excluded because a fit of either channel, unperturbed or perturbed, terminated at a bound. The split by perturbation was recorded only for the earlier run on the unaligned reference:

Table 9. Calibration coverage of the retrieved records, from calibration blocks extracted directly from the raw files: number of records n ; percentage of records outside the knot range and inside a knot gap longer than 2 h (counts from the coverage extractor; the retrieval itself has 9149 ambient records per heated channel in the budget window), for the zero-air (I_0) and reflectivity (R) references; and the longest reflectivity gap.

instrument	window	n	I_0 out of range	I_0 gap > 2 h	R out of range	R gap > 2 h	worst R gap
hot ch1/ch2	campaign	76 282	0.08 %	0.07 %	3.76 %	1.23 %	10.6 h
hot ch1/ch2	budget window 05-18..24	9 088	0.01 %	0.00 %	0.00 %	0.00 %	1.3 h
cold	campaign	43 259	0.13 %	0.34 %	2.89 %	11.03 %	11.8 h
cold	cold window 06-10..16	7 069	0.42 %	0.00 %	17.32 %	18.31 %	11.8 h

there, in hot PNs, the zero-air perturbation accounted for 121 bound-terminated records against 4 for the unperturbed fits (the categories overlap), and the hot ANs channel had none. In that run the operational fit was well inside its feasible region, and it was the structural perturbation that drove it out – which is to say that the margin between the operational solution and the edge of identifiability is smaller than the size of a reference error. This is also the ground on which Sect. 4 excludes bound-terminated records from Table 5: a fit pinned at a bound has not converged, and its response is not a measurement of anything.

What the pipeline could report instead. The optimiser distinguishes a fit stopped by the gradient from one stopped by a step limit or at a parameter bound, and Augur computes these as distinct states internally. In the delivered campaign product they were not carried alongside the retrieved amount; the current version exports them as separate columns (Sect. 2.6). Collapsing them into “converged” is what allows a bound-pinned shift to be published as a measurement.

5.3 How far a record is from its calibration

Every retrieved amount sits at some distance in time from the calibration knots that produced its reference spectra, and that distance varies over the campaign by more than an order of magnitude. The pipeline computes it – it has to, in order to interpolate – and then discards it. The retrieval interpolates the reflectivity across an 11.8 h gap exactly as it does across a one-hour gap, and returns the same uncertainty in both cases.

Coverage was measured over the whole campaign by extracting the calibration blocks directly from the raw files, independently of the retrieval, and verified against the production knot cache (157/157 hot and 164/164 cold knots at 0.000000 s) and against the alpha header’s own `ZA_count=1313`. Two distinct deficits appear, and they are not the same quantity (Table 9). For the zero-air reference the word “interpolation” is accurate: 0.08 % of hot records lie outside the knot range and 0.07 % inside a gap longer than two hours, with the nearest knot a median 922 s away. The zero-air knot set is on the corrected time axis – the production cache over 05-18..24 reproduces a re-extraction at 0.000000 s – so the I_0 column is unaffected by the displacement described in Sect. 5.4.

The cold channel’s deficit is real and is a gap problem: the existing quality gates reject 11.3 % of cold pairs against 0.24 % of hot pairs (3 of 1 264); an independent count gives 75 of 743, 10.1 %, and nothing below turns on the difference. Independent

Table 10. Change in the retrieved amounts when the heated channels are retrieved with the clock-aligned instead of the nine-hour-displaced reflectivity knots, over the records of the budget window: robust scale (rSD) and standard deviation (SD) of the change (ppb), and the rSD of the change relative to the rSD of the R term in Table 5. Σ ANs is formed with $g' = 0.9524$ as in Table 5.

series	delta rSD (ppb)	delta SD (ppb)	vs the R term of Table 5
ANs channel	0.1048	0.1627	3.0
PNs channel	0.0358	0.0563	2.8
Σ ANs	0.100	0.1418	3.7

rejection at the 11.3 % rate would leave 3.5 % of cold records in gaps longer than two hours against the 11.03 % observed, and the worst gap, 11.8 h, is eleven consecutive rejected pairs – a run that even at a 40 % independent rejection rate would be expected 0.03 times in the campaign. The rejections arrive in runs, which is the episodic structure Sect. 6.1.1 found in the reflectivity variance itself, seen from the calibration side.

600 The two deficits also explain a limit on Table 5 itself. A perturbation experiment can only be run on scans that have neighbouring knots of both reference types, so periods in which the calibration failed are absent from the budget population – and those are precisely the periods with the largest errors. In the cold channel, eleven of the thirteen zero-air blocks identified as defective in Sect. 6 fall outside the case set because no reflectivity knot exists within 2.8–5.8 h of them: a collapse of the zero-air signal takes the He/zero-air pair down with it, so the two reference failures are one event seen twice.

605 **What the pipeline could report instead.** Three numbers the interpolation already holds: the elapsed time to the nearest knot of each reference type, the length of the enclosing knot interval, and a flag for records outside the knot range altogether rather than between knots. Section 6.1 specifies them as product fields. They cost nothing to emit and they separate the 0.00 % of the budget window from the 18.31 % of the cold window, which no field in the current product distinguishes.

5.4 A reference error the fit does not see

610 Table 5 is computed on a clock-aligned reflectivity reference. The operational one was not aligned: the reflectivity knots for 2026-05-18 to 05-30 lie on a time axis that a later correction to the raw-data reader superseded, while the spectra lie on the corrected one, so every record in that span was retrieved with a reflectivity displaced by nine hours. The budget window of Sect. 4 falls inside it. The zero-air reference is unaffected – its knots were already on the corrected axis, verified to 0.000000 s – and so is the cold channel, which the correction does not apply to.

615 This makes it possible to measure directly what using a wrong reference costs, and what the fit report says about it. The retrieval was run over all 9 132 records of the window common to both heated channels twice, changing nothing but the knot time base (Table 10).

 The displacement moves the alkyl nitrate product by 0.100 ppb: between three and four times the reflectivity term of the budget, and 1.44 times the whole of it. It is the largest term we can quantify within the retrieval; the multiplicative scale

Table 11. What the fit report shows for the displacement of Table 10: RMS change of the extinction spectrum (cm^{-1}), ratio of the residual and of the reported σ between the two retrievals, change in the retrieved amount in units of the reported σ , and change in the fitted shift.

	α RMS change	residual ratio	reported σ ratio	$ \delta c /\sigma$	$ \delta \text{shift} $
ANs	3.37×10^{-9}	1.0020	1.0022	1.99	0.0053 px
PNs	1.53×10^{-8}	1.0016	1.0016	0.55	0.0065 px

620 discrepancy of Sect. 3.6 is larger, but lies outside it. It is not in Table 5 because Table 5 measures what a reference’s own uncertainty costs, not what using the wrong reference costs.

What the fit reports about it (Table 11) is the point of the section.

The extinction spectrum changes by an amount comparable to the single-scan noise floor of 6.16×10^{-9} . The residual and the reported uncertainty change by the same two parts in a thousand, and the agreement is not a coincidence: across the campaign
625 the reported uncertainty is a near-constant multiple of the residual RMS ($r = +0.999$; the ratio moves by 2.8 % between its 5th and 95th percentiles, Sect. 7.3). A fit report that shows an unchanged residual and an unchanged uncertainty is not offering two independent witnesses to its own quality. It is offering one, twice. The fitted wavelength shift moves by 0.005 px, where the reference perturbations of Table 5 move it by 0.243 px (median) – about forty times less – so the error is neither absorbed by the nonlinear parameters nor revealed by the residual. The retrieved concentration of the 300 °C channel moves by two
630 reported standard deviations. Every quantity a fit report offers as evidence of quality is unchanged; the answer is not.

Correcting it did not simply reduce the budget, and the way it did not is informative. The robust scale of the Σ ANs budget fell by 7.0 % (0.0745 to 0.0693 ppb) while its standard deviation rose by 4.5 % (0.1183 to 0.1236), so the total uncertainty relative to what is reported widened at both ends, from 1.70–2.41 to 1.63–2.50 times against the same fit-reported sigma (Sect. 4.4). Two things were working in opposite directions. Coverage improved, from 8505 to 9034 of the window’s 9149
635 ambient records (93.0 to 98.7 %), because the misaligned knots had been falling outside the one-hour window that admits a record to the perturbation experiment. And the correlation between the two channels’ joint responses fell from +0.412 to +0.146: because Σ ANs is a channel difference, a common-mode error partly cancels in it, and the pre-correction cancellation was an artefact of an error the two channels shared. The contaminated product was flattering itself in two ways at once, and the budget measured on it inherited both.

640 The cost is not uniform in time, and not in the direction we predicted before measuring it. Across the boundary where the clock was corrected, the affected side gives 0.0622 ppb against 0.0000 ppb on the unaffected side – the latter a check that the realignment touched only the intended knots – so the displacement costs less near the boundary than in the budget window. What matters is not proximity to the boundary but how fast the reflectivity is moving at the time, and it moves fastest early in the deployment. The budget-window figure of 0.100 ppb is the value to quote; 0.0622 ppb is range information, not a worst
645 case.

What the pipeline could report instead. The failure here is not that the heated-channel clock ran on local time until 29 May – a property of the raw record, identified against an external time reference and corrected in the reader. It is that a derived cache

went stale and nothing noticed. The reflectivity file was written on 2026-07-10; the correction to the reader that invalidated it was committed on 2026-09-19; the retrieval read both without comparing them. A pipeline that records, for each derived reference, the version of the code and the raw files it was built from can detect this condition mechanically, and it is the only failure in this paper that admits a complete remedy rather than a better report. Section 6.1 specifies the fields.

6 What should be reported, and what should be changed

Section 4 established that the fit-reported uncertainty omits terms that are several times larger than itself, and Sect. 5 that convergence does not imply identifiability. Neither finding requires a new retrieval algorithm to act on. This section states what a DOAS product should carry instead of a single uncertainty, and what in the measurement schedule is worth changing. Of the four recommendations, two are implemented in Augur and evaluated against the campaign: the knot gate (Sect. 6.3) and the coverage fields (Sect. 6.4). The reporting convention of Sect. 6.1 is implemented in part – the termination state, the distance to the nearest knot and the identity of the reference file are exported, while the build time and time convention of each reference are specified for implementation – and the question of block length in Sect. 6.2 is left open by the measurements reported here. Section 6.5 names the errors that none of these measures reaches.

6.1 Three numbers per retrieval, not one

Every retrieved amount is reported with

1. the **fit uncertainty**, from the covariance of the projected problem (Sect. 2.5) – what the residual supports, and nothing more;
2. the **structural term**, evaluated for that scan’s own calibration context rather than as a campaign constant; and
3. a **context flag** recording the state of the references the scan depends on, in the specific form set out below.

The third item is the one that has to be specified rather than gestured at, because its value is entirely in which fields it carries. We report, per record and per reference type (zero air, reflectivity), the fields of Table 12.

The last field is there because of Sect. 5.4, and it is worth being exact about what it would and would not have caught. The heated-channel clock ran on local time until 29 May and on UTC thereafter, so for twelve days of the campaign the raw time stamps were nine hours off the convention of the time axis; that is a property of the raw record, and no provenance field detects it. It was detected the way such things are detected – against an external time reference – and a correction was added to the raw-data reader two months after the deployment ended.

What the field catches is the failure that followed. The reflectivity knots had been computed before the correction existed, and nothing rebuilt them afterwards. From that moment the retrieval was combining spectra on the corrected time axis with a reference on the superseded one, and the twelve days of product made from that combination are wrong by more than their whole structural budget. This is not a clock problem; it is a staleness problem, and it is general. Any correction applied upstream

Table 12. Fields of the context flag, reported per record and per reference type (zero air, reflectivity), and the reason for each.

field	why
seconds to the nearest knot, with the side (before/after)	the side distinguishes interpolation from extrapolation
length of the bracketing interval	an eleven-hour gap and a fifteen-minute one are not the same interpolation
in-range flag	extrapolation is a different operation and should be nameable in a filter
number of adjacent calibration blocks rejected by the quality gates	the gates' cost, which Sect. 6.4 argues must be reported wherever the gates are applied
identifier of the reference file, and when it was built	two products made from different calibration vintages are not comparable
the time convention the reference was built under	a correction applied upstream makes every earlier derived product stale

invalidates every derived product built before it, and a pipeline in which derived products do not record the convention they were built under cannot tell that it has happened.

680 The mismatch was not silent in the data. It surfaced as a nine-hour run of out-of-range records at the very start of the deployment – the first zero-air block at 02:42, the first reflectivity knot at 11:40 – and 42.3 % of that day extrapolated. The distance and in-range fields would have carried that anomaly into the product on the first run after the correction, rather than two months later. They would not have explained it: identifying the cause needed the build time of the reflectivity file and the convention it was built under. Detection and diagnosis are separate functions and need separate fields.

685 There is also a cheaper protection than reporting, and it belongs in the same subsection because it is the same information used earlier. A reference carrying a time-convention tag can be checked against the tag on the spectra at load time, and a mismatch refused rather than reported. Reporting turns a silent error into a visible one; an assertion prevents it. We recommend the assertion where the tags exist and the report in every case, because the report is also what makes the error interpretable a year later, when the person reading the product is not the person who ran it.

690 The second of these cannot be obtained cheaply, and it is worth saying why. The amplification factor $I_0/(I_0 - I)$, which governs how an error in the zero-air reference enters the extinction, varies by a factor of forty between records and is available without any fitting; it is a natural candidate for a closed-form structural term. It does not work. Regressed against the measured response it explains between 0.0 % and 9 % of the variance, with the wrong sign, because the polynomial background absorbs the smooth part of the perturbation and what reaches the gas columns is set by shape rather than amplitude. Normalising the
695 response by the retrieved amount does not help either: the robust coefficient of variation is 0.80–1.09 in relative terms against 0.83–1.11 in absolute. The structural term has to be obtained by running the perturbation. That is available in post-processing and not in a real-time monitor, and the two implementations differ accordingly.

 The second and third exist because the first is conditional. A fit uncertainty is a statement about the residual of a model assumed correct; it is not a statement about the inputs to that model. Separating the three keeps that distinction visible to
700 whoever uses the product, instead of resolving it silently into one number that is too small.

Table 13. Concentration in time of each structural term over the hours of the budget window (clock-aligned basis of Table 5): shape statistic s ; number (and fraction) of hours holding 90 % of the term’s variance; the number of contiguous runs those hours form, with the number expected for the same count of hours drawn at random; share of the variance on the standard-deviation basis; and the remedy the shape indicates.

term	s	hours holding 90 % of the variance	contiguous runs (ran- dom expectation)	share of variance (SD)	remedy
I_0	0.73	63 (41 %)	24 (38)	45 %	longer calibration blocks (untested, Sect. 6.2)
R	0.32	26 (17 %)	12 (22)	52 %	detect and flag intervals (Sect. 6.3)
etalon f	0.98	118 (76 %)	26 (29)	3 %	fit f per scan, or model it

6.1.1 Which remedy a term needs is readable from its shape

The three structural terms call for three different actions, and the shape statistic s of Sect. 4.3 says which. A term with $s \approx 1$ is Gaussian: its error is spread evenly over all records and can only be reduced by a better model. A term with $s \approx 0.7$ –1 is dominated by measurement noise on the references and responds to averaging. A term with $s \ll 0.7$ is concentrated in a few records, where averaging is useless and detection is everything.

This is not an interpretation of s but a prediction from it, and it can be checked. Ordering the 155 hours of the budget window by their contribution to each term’s variance gives Table 13.

The ordering by s is the ordering by concentration, and the two-hour interval from 23:00 UTC on 18 May to 01:00 UTC on 19 May alone holds 37 % of the reflectivity term’s variance. The contiguous-run counts separate concentration from clumping: the reflectivity hours form far fewer runs than the same number of hours drawn at random, so they are episodes rather than scattered outliers and a flag can address them. The etalon hours are indistinguishable from random, which is the signature of a term no flag will help.

Read on the robust scale the ordering of the first two reverses – I_0 carries 74 % of the single-term quadrature sum and R only 17 % – which is why the two scales are reported side by side in Table 5. A term that is small for a typical record and largest in total is exactly the one a campaign-mean uncertainty hides.

6.2 Calibration blocks: interval length and scans per block

The natural response to an interpolation error is to interpolate over shorter intervals. How much that buys depends on the channel. Tripling the reflectivity interval from 2 to 6 h was measured to increase the interpolation error from 0.87 % to 1.00 % in hot PNs and from 2.05 % to 2.73 % in hot ANs. In the 180 °C channel the error is set mostly by the noise on the knots, and a shorter interval would buy little; in the 300 °C channel the interval carries curvature (Table 4), and shorter intervals would help.

The other lever is the number of scans averaged into each calibration block, and the evidence on it is not conclusive (Appendix C). Within the operational block ($N = 34$ scans, $T = 32$ s), doubling the number of scans averaged reduces the scatter of contiguous sub-blocks by 7 % in the ANs channel, against the 29 % that independent scans would give; but the range that
725 can be measured this way lies entirely below the operational block length, so it is mildly unfavourable rather than decisive, and the measurement that would settle it – operating one longer block – has not been made. Even with white averaging the return would be modest: the zero-air term carries 45 % of the Σ ANs variance, so a fourfold block, which would raise the calibration duty from 1.20 to 3.89 %, would lower the structural budget from 0.1236 to 0.1008 ppb (18.5 %), whereas flagging the few defective reflectivity intervals would lower it to 0.0901 ppb. That ordering – flagging before averaging – holds whatever the block
730 scaling turns out to be, because the reflectivity term is concentrated in a few episodes and the zero-air term is not (Sect. 6.1.1).

6.3 Gating defective knots

Not all calibration blocks are usable, and the bad ones are few. Ordering the cold-channel zero-air blocks by their leave-one-out influence, the worst single block carries 28 % of the total variance, the worst three carry 76 %, and the worst five carry 97 %. A structural term computed over all blocks is therefore a statement about a handful of them.
735 They are identifiable before any fit is attempted. A single observable – the block’s light level relative to the campaign median – with the threshold $I/\tilde{I} < 0.5$ selects 13 of the 38 blocks whose influence exceeds 2 %, but those 13 include nine of the worst ten and account for 99.9 % of the influence variance. The 25 blocks it misses are all in the 2–3 % range. The threshold excludes 8 % of blocks.

Classification accuracy is the wrong figure of merit here and would have rejected this threshold: 13 of 38 is a poor hit rate
740 and a good filter. What matters is the variance removed, and a single observable removes essentially all of it. No multi-indicator scheme is needed.

The asymmetry with the reflectivity is worth recording, because it explains why only one of the two references needed this. Reflectivity knots already pass three quality gates – a minimum fraction of valid pixels, an acceptance band on the retrieved effective path length, and limits on the calibration residual – so a defective helium or zero-air pair is rejected before it becomes
745 a knot. The zero-air reference had no equivalent: every block became a knot regardless of how much light reached the detector. The gate described here closes that gap. It is implemented with the threshold expressed as a fraction of the run’s own median block intensity rather than as an absolute count, so it transfers between instruments, and it is disabled by default so that reprocessing an existing product is an explicit choice rather than a silent change.

6.4 Coverage: say when the reference is far away

750 Two things are missing, and conflating them produces the wrong prescription. One is *extrapolation*: records outside the range spanned by the reflectivity knots. The other is a *gap*: records between two knots more than two hours apart. The hot channel has 3.76 % of the first and 1.23 % of the second over the campaign; the cold channel 2.89 % and 11.03 %. Zero air has neither (0.08 % and 0.07 %), so the constraint is entirely on the reflectivity.

Two causes lie behind the deficit that survives clock alignment, and they cost different things to remove. The first is book-keeping and the cheapest fix in this paper: reflectivity generation stopped at 07-09, which produces the hot channel's trailing extrapolation and is removed by re-running over files that already exist. The second is clustered rejection of calibration pairs, 11.3 % of cold pairs, which explains the cold gaps (11.03 % over the campaign and 18.31 % in the cold window); its remedy is to restore the light level or, failing that, redundancy. Only this second cause is about how often the instrument calibrates, and even there the average rate is not the problem. Independent rejection at the measured rate would leave 3.5 % of cold records in gaps longer than two hours, against 11.03 % observed. The rejections arrive in runs: the worst cold gap, 11.8 h, is eleven consecutive rejected pairs, which even at a 40 % independent rejection rate would be expected 0.03 times in the campaign. This is the episodic structure Sect. 6.1.1 found in the reflectivity variance itself – 26 of 155 hours carrying 90 % of it, in 12 contiguous runs where 22 would be expected at random – seen from the calibration side.

It also locates the cost of quality control. The three gates the reflectivity already passes cost the hot channel 0.24 % of its pairs and the cold channel 11.3 %; in the cold channel the gates are, correctly, the proximate cause of the coverage deficit. This cuts against the argument of Sect. 6.3, which asks for a gate that zero air does not yet have: gates remove variance and they remove coverage, and a pipeline that gates without also reporting what the gating cost has moved the error somewhere it is harder to see. The context flag of Sect. 6.1 closes that loop. The same caution applies to the budget: the reflectivity contribution of Sect. 4 is measured by removing a knot and re-interpolating, so its magnitude scales with how far apart the knots already are, and a sparse reference presents itself as a dominant term partly because it is sparse – which is why the remedy belongs to the schedule and to reprocessing rather than to the retrieval. Records in gaps are not covered by the budget at all, since a perturbation experiment needs a knot to perturb; they are reported with the context flag set and excluded from any statistic that assumes a characterised uncertainty.

In the heated channels zero air is measured every hour and helium every three, and each reflectivity knot pairs a zero-air block with the most recent helium block that passed the quality gates (Appendix A, Table A2), so the knots are hourly while their helium half is up to three hours old, and older when a helium block is rejected. A rejected zero-air/helium pair removes one hourly knot and leaves a two-hour gap; a rejected helium block removes no knot but makes the following knots use helium measured up to six hours earlier. Matching the helium cadence to the zero-air cadence would bring the total calibration duty from 1.20 % to 1.79 % and make the helium half of every knot contemporaneous. Whether it would also shorten the gaps depends on why pairs are rejected, which in the cold channel is mainly the light level. The case for it is redundancy against runs of rejection and contemporaneity, not precision: in the 180 °C channel the reflectivity error depends only weakly on interval length (Sect. 6.2), so more frequent helium would not make a well-covered record much more accurate. At 1.2 % of the measurement time, calibration is the binding constraint on none of these remedies.

6.5 Errors that neither the fit report nor the proposed fields detect

The convention of Sect. 6.1 makes visible what the pipeline already knows. It cannot make visible what the pipeline does not know, and two such errors appear in this record.

The first is a multiplicative gain error. Against a co-located NIER analyser the heated-channel NO₂ reads low by a factor of about 0.6 while the ambient-temperature channel agrees (Sect. 3.6). A gain leaves the residual, the reported uncertainty, the termination state and every distance and provenance field unchanged, and the zero-air test of Sect. 7.2 is blind to it because a gain applied to zero is zero. Only a comparison with an independent measurement of the same quantity – a reference analyser, a cylinder, or a channel that should agree – reveals it, and such a comparison does not by itself identify the cause.

The second is the raw light level. All six ambient-temperature excursions of Sect. 7.3 coincide with spectra that fell below 30 % of their usual count at 460 nm; the residual finds them, and the reference context is abnormal in five of them, but none of the proposed fields names the cause. The light level of each spectrum, and the helium and zero-air intensities of each calibration block rather than only their ratio, are known when they are recorded and could travel with the product as further context; we have not evaluated them for the heated channels.

7 Field demonstration

7.1 Instrument and deployment

The field case is a thermal-dissociation broadband cavity-enhanced absorption spectrometer with two heated and one ambient-temperature cavity, of the design described by Min et al. (2016) and Nam et al. (2022). Ambient air is split between a cell held at 180 °C, in which peroxy nitrates dissociate, and one at 300 °C, in which alkyl nitrates dissociate as well; a third cell at ambient temperature measures NO₂ directly. The three differ in what has been converted to NO₂ upstream rather than in how the spectrum is formed, and the sums of peroxy and alkyl nitrates follow as channel differences, Σ ANs with the inter-channel factor g' of Sect. 4.4. The retrieval configuration of each channel – window, baseline, references, shift and squeeze settings, wavelength calibration, line shape, cavity constants and calibration schedule – is given in Appendix A.

The instrument was deployed at Suncheon Shindae (SDD) from 2026-05-18 to 07-10 as part of a multi-institution intensive campaign. Two properties belong to this deployment rather than to the configuration: the median fitted shift in the 300 °C channel is –6.12 px, inside its bound of –10 px, and the field numbers of this paper were checked against a reprocessing with a corrected pressure-sensor pairing, which changes the channel NO₂ by ± 0.25 %, and are reported as first computed. The instrument itself, its inlet and its conversion efficiencies are characterised elsewhere.

7.2 Noise floor of the product record

The quantity the product reports is a channel difference, so its noise floor has to be measured on the difference and not assembled from the channels. The two hot channels share a light source, a cavity and a clock, and their zero-air blocks coincide in time, so 155 blocks pair exactly and the difference can be formed record by record (Table 14). Assembling the floor from the channels in quadrature would assume an independence that there is no reason to expect.

Table 14. Noise floor of the Σ ANs product, measured on the paired zero-air blocks of the two heated channels: 1σ , 3σ detection limit, median at a true zero, and the ratio of the measured 1σ to the uncertainty column shipped with the product. Standard-deviation values are approximate, recombined from the channel values with $g' = 0.9524$; robust-scale values are from the earlier evaluation with a channel factor of 0.82 and were not recomputed.

	robust scale	standard deviation
measured 1σ	0.0676	≈ 0.086 ppb
detection limit, 3σ	0.203	≈ 0.257 ppb
median value at a true zero	+0.0054 ppb	+0.0054 ppb
against the product-column 0.0806 ppb	0.84	≈ 1.06

The channels turn out to be nearly uncorrelated, $r = -0.093$, so the quadrature assembly would have come within about 4 % of the measured value. That is a result of the measurement rather than a licence to skip it: the same comparison on an instrument whose channels shared a drift would not be so forgiving, and nothing in the fit report distinguishes the two cases.

820 The median at a true zero is +0.0054 ppb, and it is the same at the channel level and in the difference. A product assembled through zero-air interpolation, a Rayleigh correction, a time-interpolated reflectivity, a spectral fit and a channel subtraction returns zero when the truth is zero, to a twentieth of its own detection limit. This closes the additive chain end to end and belongs with the verification of Sect. 3.1 as much as here; a zero test is blind to gain errors, such as the scale discrepancy of Sect. 3.6.

825 Read against the uncertainty column shipped with the product, the product’s noise floor is roughly covered: 0.84 to about 1.06 times that column’s 0.0806 ppb, depending on the scale. This is not in tension with Sect. 4, and the distinction matters enough to state. What is measured here is a zero-air floor – the same references, minutes apart, no atmosphere and no elapsed time for a reference to go stale. What Sect. 4 measures is what happens when a reference is wrong, and that is a different question with a different answer: 1.32 to 1.83 times the same product-column figure of 0.0806 ppb (Sect. 4.4). (Against the fit-propagated σ of 0.0541 ppb, the denominator of Sect. 4.4, the total is 1.63 to 2.50 times that value, and in 46.1 % of the 830 8832 budget records the structural change exceeds that record’s own fit-propagated σ .) A reported uncertainty can describe the photon noise of a product faithfully and still say nothing about the references the product was built from.

835 The reported uncertainty column of the product is not the quadrature of the channel columns – it is 1.48 times larger, 0.0806 ppb against 0.0543 – because it is not a measurement uncertainty. It is the inter-channel gain systematic described in Sect. 4.4, which scales with the NO_2 amount, and for that reason the ratios of Sect. 4 are taken against the fit-propagated sigma rather than against this column.

7.3 The reporting convention in operation

Applied retrospectively to this deployment, in the heated and the ambient-temperature channels alike, the quantities Sect. 6.1 asks for separate what the fit report catches from what it does not (Fig. 6); the provenance fields among them are specified, not yet implemented.

840 One episode is not caught by the fit report at all: the reference-convention displacement of Sect. 5.4. From 18 to 30 May the heated channels were retrieved with a reflectivity displaced by nine hours; the alkyl nitrate product moved by 0.100 ppb while the residual and the reported uncertainty changed by two parts in a thousand. Across the deployment the reported uncertainty of the ANs channel is a near-constant multiple of the residual RMS ($r = +0.999$ over all 74 667 records; the ratio varies by 2.8 % between its 5th and 95th percentiles), so the two are one witness, not two. Among the proposed fields, the distance and
845 in-range fields flag these records (Sect. 6.1) and only the provenance fields would name the cause.

The ambient-temperature channel supplies the opposite case. Its six excursions below -5 ppb (19 hourly medians between 19 May and 2 June) are all visible in the fit itself: the residual RMS is 8 to 600 times its campaign median and the reduced chi-square 5.6 to 91, against 1.2 in ordinary hours. Five of the six coincide with an abnormal reference context – the bracketing reflectivity knots are 2.6 to 9.0 h apart against 1 h nominal, or one of them departs from its running median by 66 to 300 %
850 against a typical 10 % – so the distance-to-reference and knot-validity fields explain them, but are not needed to find them. The sixth, four hours on 2 June, has an ordinary reference context and is identified by the residual alone. All six share a proximate cause that none of these fields names: the raw light level. In each, the ambient or the zero-air spectrum, or both, fell below 30 % of its usual count at 460 nm, in four of them to within a few hundred counts of the detector dark level; on 2 June only the ambient path lost light (22 % of the zero-air level), which is why the reference stayed ordinary (Fig. 6). The misfit threshold
855 ($\chi^2 > 1.5$) is itself signal-dependent (Sect. 4.1); these excursions exceed it by an order of magnitude or more.

In the heated channels no hourly median falls below -0.5 ppb over the deployment. The termination-state field flags one episode: for eleven minutes on 21 June the 180 °C channel returned its shift at the $+5$ px bound, and the residual is raised over the same records (reduced chi-square 4.7 to 5.5). Here the residual signals that something is wrong and the termination state says what.

860 In each of these cases the fit either reports an ordinary residual and uncertainty for a displaced amount, or reports an abnormal one without saying why. For the displacement and for the bound episode, one of the context quantities – distance to the nearest reference, termination state, or reference provenance – supplies what is missing. For the cold excursions the proximate cause, the raw light level, is not among the proposed fields, and the heated-channel NO₂ scale of Sect. 3.6 is invisible to all of them (Sect. 6.5). The fields narrow the gap between what a fit reports and what is known about the record; they do not close it.

865 8 Conclusions

We have presented Augur, an open retrieval posed in the extinction domain as a separable nonlinear least-squares problem and solved by variable projection, configured by instrument and campaign profiles rather than written for one instrument, and demonstrated its diagnostics on one field case: a thermal-dissociation broadband cavity-enhanced absorption spectrometer

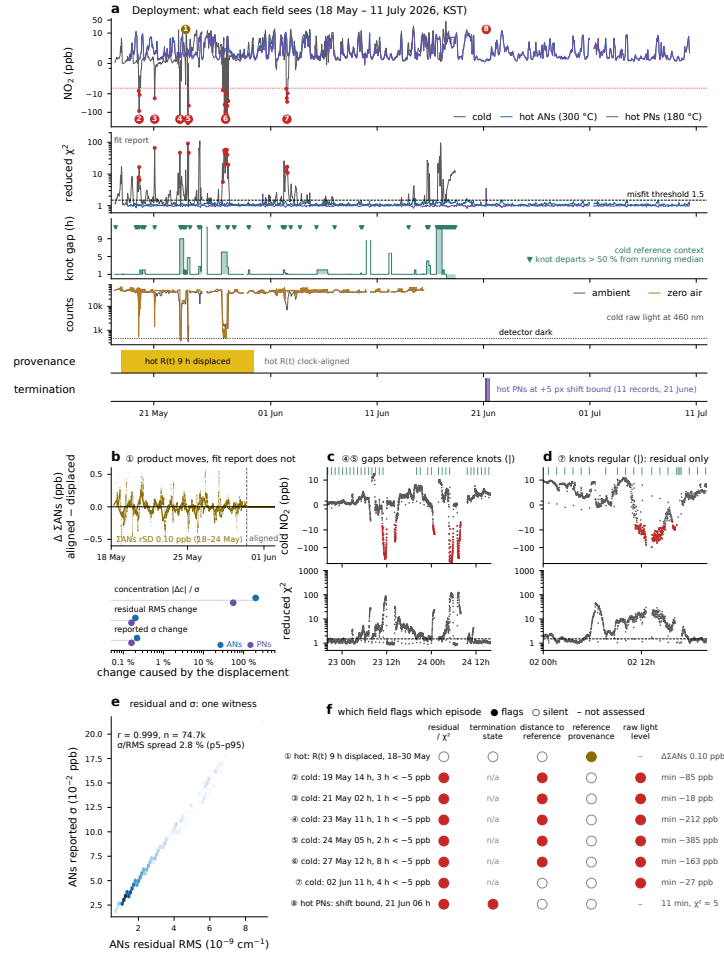


Figure 6. What each reporting field flags during the deployment (18 May – 11 July 2026, KST). **(a)** From top: hourly median NO₂ of the three configurations (symmetric-log axis; red: cold hours below –5 ppb); hourly median reduced χ^2 with the misfit threshold of 1.5; for the cold channel, the interval between the reflectivity knots that bracket each record (triangles: a knot departing by more than 50 % from its running median); the raw cold-channel light at 460 nm in ambient and zero-air spectra, with the detector dark level; the provenance of the heated-channel reflectivity; and records that terminated at a shift bound. Numbered badges mark the episodes of panel f. **(b)** Episode ①: the change in ΣANs when the heated channels are retrieved with the clock-aligned instead of the 9 h displaced reflectivity (points: records; line: 6 h median; dashed: end of the displaced span). Below, for 18–24 May, the change in concentration expressed in units of its reported σ , against the change in residual RMS and in reported σ . **(c)** Episodes ④ and ⑤: cold NO₂ and reduced χ^2 per record, with reflectivity knot times as ticks; the excursions fall in 9.0 h and 4.7 h gaps. **(d)** Episode ⑦: the same for 2 June, where the knots are regular and only the residual flags the records. **(e)** Reported σ against residual RMS for every hot ANs record. **(f)** For each episode, whether the residual ($\chi^2 > 1.5$), the termination state, the distance to the nearest reference (knot interval > 2 h or knot departure > 50 %), the reference provenance, or the raw light level (ambient or zero-air count below 30 % of its usual value) flags it; the light level was not assessed for the heated channels (–); the cold configuration has fixed shift and squeeze, so its termination state carries no bound information (n/a).

with two heated and one ambient-temperature cavity. The formulation is not the contribution. The implementation makes three things cheap: substituting a calibration reference and re-running the retrieval without reprocessing the campaign, evaluating the projected objective anywhere in the nonlinear parameters, and carrying to the product the state in which the optimiser stopped and the distance to each calibration reference; recording each reference's build time and time convention is specified here but not yet implemented. That turns reliability questions from studies into routine measurements.

The code solves the problem it states: on noise-free synthetic spectra with integer shifts the recovery is at the level of the linear algebra, the analytic projected Jacobian agrees with finite differences, and Augur and QDOAS agree to within the single-scan reported uncertainty; on field spectra the two codes agree in NO₂ with $r^2 \geq 0.990$ but differ in the heated-channel scale by 4–6 %. On the deployment the full chain returns +0.0054 ppb at a true zero, a twentieth of the product's three-sigma detection limit of 0.203 to about 0.257 ppb – a test of the additive chain only.

The uncertainty a fit reports is correct where its model is: with white noise, a complete forward model and a well-identified shift the reported sigma tracks the synthetic scatter to within about ten per cent, and with a free shift the ratio reaches 1.32. Against the calibration chain it does not. For the channel-difference product, perturbing the zero-air reference, the reflectivity and the etalon as the operational pipeline builds them gives, in the best-calibrated week, a structural contribution of 1.28 (robust scale) to 2.29 (standard deviation) times the reported sigma, and a total uncertainty 1.63 to 2.50 times what the fit reports for the same 60 s records. None of the calibration-chain terms enlarges the residual. The magnitudes are this instrument's; the mechanism, an intermittently measured reference interpolated to every record, is general.

A second class of shortfall needs no new measurement, because the pipeline already holds the information. With the cold shift bounded to ± 1 px in a sensitivity run, every base fit of the ambient-temperature channel returned its shift at the bound, and in the budget propagation 187 of 9034 records were excluded because a fit, unperturbed or perturbed, terminated at a bound; all were recorded as converged. The rate depends on the step limit (Fig. 1b) and differs between implementations (Sect. 5.1), but the state is computed and then discarded. In one window the projected objective varies by about one per cent across four pixels of shift, in another over five orders of magnitude, and both are reported as converged. And a reflectivity reference on a superseded time convention displaced the alkyl nitrates by 1.44 times the structural budget while moving the residual and the reported uncertainty by two parts in a thousand. Neither the fit report nor the proposed fields see the largest error in this record (Sect. 6.5): against a co-located NIER analyser the heated-channel NO₂ reads low by a factor of about 0.6 while the ambient-temperature channel agrees, a multiplicative discrepancy whose cause is unresolved (Sect. 3.6).

We therefore propose that a retrieval report three numbers rather than one: the fit uncertainty; the structural term for that record's own calibration context rather than a campaign constant; and a context flag carrying the distance to the nearest knot and its side, the bracketing-interval length, an in-range flag, the number of adjacent blocks the quality gates rejected, and each reference's identity, build time and time convention. Every implementation already computes all of these except the build time and time convention of a reference, which must be recorded when the reference is built; no change of algorithm is needed. Two operational points follow. Where the interpolation error is set by knot noise, as in the 180 °C channel, averaging more scans per calibration block rather than running blocks more often is the lever to test; the within-block evidence of Sect. 6.2 is

unfavourable and the decisive measurement has not been made. And quality gates should be reported with the coverage they cost, because they remove coverage as well as variance (Sect. 6.4).

905 The evidence is one field case, one instrument over one campaign. The code is not tied to that instrument (Appendix A), but its behaviour on other configurations is not shown here. Two things should carry beyond it: the reporting convention, which asks for quantities every implementation has or can record when a reference is built, and the synthetic benchmark of 261 cases, which contains no retrieval code and no field data and so favours no implementation – though its cross sections (convolved with this instrument’s line shape) and windows make it independent of the retrieval code, not of the instrument. Extending it
910 across line shapes, windows and pixel grids would separate what is general from what is ours; we release it so that others can do so.

Three directions follow. The wavelength shift belongs to the spectrograph, not the air, so fitting one shift jointly across the spectra of a calibration block – a separable problem with common nonlinear parameters – could remove the indeterminacy this paper only diagnoses (Sect. 5.1); a laboratory spectrum with a few ppb of added NO₂ already locates the ambient-temperature
915 channel’s shift to a tenth of a pixel (Sect. 5.1), so existing calibration blocks would serve if the absorber were supplied. And treating the zero-air reference and the reflectivity as measured quantities with their own covariance, not exact inputs, would propagate the structural term inside the fit rather than afterwards (Sect. 4.6).

Third, the raw light level of each spectrum – the proximate cause of all six ambient-temperature excursions of Sect. 7.3 – and the helium and zero-air intensities of each calibration block, not only their ratio, should travel with the product (Sect. 6.5),
920 so that changes in the light reaching the detector can be seen where they happen instead of being inferred afterwards from the product.

Code and data availability. Augur is free software released under the MIT licence. The version used in this paper, v0.2.0, is archived at Zenodo ([https://doi.org/\[TBD-Z1\]](https://doi.org/[TBD-Z1])) and developed at <https://github.com/HwanKyungLee/CAESAR>. The archive contains the analysis code, the test suite referred to in Sect. 3, the scripts that produce the figures of this paper, and derived calibration constants (wavelength-calibration
925 polynomials, instrument line-shape widths and instrument-convolved reference cross sections). The original laboratory cross sections are not redistributed; they are available from their providers: NO₂ (Vandaele et al., 2002), CHOCHO (Volkamer et al., 2005), O₄ (Thalman and Volkamer, 2013), and H₂O line data from HITRAN (Gordon et al., 2022).

The synthetic benchmark of Sect. 3.4 – 261 spectra with known truth, their generator, the reference implementation and its scorecard – contains no field data and is archived separately ([https://doi.org/\[TBD-Z2\]](https://doi.org/[TBD-Z2])). Figure 3 can be reproduced from the two archives alone.

930 Raw instrument data and the field time series used in Sects. 4 to 7 (60 s extinction spectra, zero-air and reflectivity knot tables, and retrieved NO₂ amounts, 18 May to 11 July 2026) are not publicly distributed; they are available from the corresponding author on reasonable request.

Appendix A: Instrument and retrieval configuration

Augur reads the layout of the raw files – which columns hold which channel’s spectrum, which state flag marks an ambient, zero-air or helium scan, and where the housekeeping values sit – from a JSON instrument and campaign profile, and the fit

Table A1. Operational retrieval configuration of the three channels. “Linked” parameters take the value of the NO₂ parameter. The squeeze is stored as the departure from unity of s_i in Eq. (3), so 0 means no stretch. Dispersion is the median over the fitting window of the wavelength-calibration file; the line-shape width of the heated channels is the value recorded in that file (mean of four Hg lines), that of the ambient-temperature channel is the pair of measurements discussed in Sect. 4.1.

	300 °C	180 °C	ambient temperature
label in Sects. 3–6	hot ANs	hot PNs	cold
raw spectrum block	2053	4101	(cold file)
fitting window (nm)	429.5–461.9	444.1–470.6	438.4–475.8
baseline	Chebyshev, order 4	Chebyshev, order 3	Chebyshev, order 4
absorbers fitted	NO ₂ , CHOCHO, H ₂ O	NO ₂ , CHOCHO, H ₂ O	NO ₂ , CHOCHO, H ₂ O
NO ₂ shift (px)	bounded, −10 to +0.5	bounded, −5 to +5	fixed, −0.5
NO ₂ squeeze, $s - 1$	bounded, ±0.005	bounded, ±0.005	fixed, 0
CHOCHO, H ₂ O shift and squeeze	linked	linked	linked
cross-section temperature coefficient	0 (not applied)	0 (not applied)	0 (not applied)
κ_i			
negative amounts	allowed	allowed	allowed
optimiser step limit	on (0.5)	on (0.5)	on (0.5)
wavelength calibration	Hg, 19 June 2026, quadratic	Hg, 19 June 2026, quadratic	Hg (four lines), 23 May 2026, quadratic
dispersion (nm px ^{−1})	0.0484	0.0481	0.0482
line-shape FWHM (nm)	0.681	0.683	0.765–0.785

935 configuration from a separate fit-setting file. The retrieval logic keys on the structure these files declare, not on channel names, so the same code is configured for another instrument or deployment by writing new files rather than by changing it. This appendix collects the configuration of the field case of Sect. 7 in one place. Table A1 is transcribed from the operational fit-setting file of the campaign and the wavelength-calibration files it names; Table 1 is its short form.

References. The cross sections are NO₂ (Vandaele et al., 2002), CHOCHO (Volkamer et al., 2005) and H₂O from HITRAN
940 line data (Gordon et al., 2022), each convolved with the line shape of its channel before the fit; O₄ is not fitted in the operational configuration. No robust weighting, regularisation or non-negativity constraint is applied (Sect. 2.8).

Wavelength calibration. Each channel has its own Hg-lamp calibration with a quadratic pixel-to-wavelength polynomial. Two calibrations of the 300 °C spectral region exist, one per region of interest on the detector; the operational fit setting uses the second, the two differ by about 0.6 px (0.034 nm), and which of the two belongs to block 2053 is not settled. The difference
945 is within the shift range of that channel and is absorbed by the fitted shift.

Table A2. Calibration schedule of the heated channels, with block counts for the budget window (18–24 May): number of blocks, scans and duration per block, cadence and duty cycle, for zero air and helium.

	blocks	per block	cadence	duty
zero air	157	34 scans / 32 s	1.00 h	0.90 %
helium	54	34 scans / 32 s	3.00 h	0.31 %
				1.20 %

Temperature and pressure. Sample temperature and pressure are recorded with every spectrum and enter the Rayleigh correction of Eq. (1) and the conversion from number density to mixing ratio; their interpolation contributes below 0.1 % (Sect. 4.1). The temperature scaling $\tau_i(T)$ of Eq. (2) is not used ($\kappa_i = 0$).

Cavity. All three cavities have $d = 51.8$ cm and $R_L = 1.0$. R_L is assumed, not measured; since $\alpha \propto R_L$ it enters as a multiplicative factor (Sect. 3.6).

Calibration references. Zero-air and helium blocks are identical, 34 scans over 32 s; zero air is measured every hour and helium every three hours (Table A2). The zero-air reference I_0 is interpolated to the time of each spectrum from the neighbouring zero-air blocks, and the etalon frequency is held fixed (Sect. 4.1). Each reflectivity knot is formed from one zero-air block and the most recent helium block that passed the quality gates, so the knots of the heated channels are hourly –
 1261 over the deployment at a median spacing of 1.00 h – while the helium half of a knot is up to three hours old; $R(\lambda, t)$ is the monotone piecewise-cubic interpolation of Sect. 2.4 through these knots, after the knot gate of Sect. 6.3.

Appendix B: The projected Jacobian, column scaling and the joint covariance

B1 Setup

Write the forward model of Sect. 2.1 for one spectrum as

$$\alpha = \Phi(\theta)c + \varepsilon,$$

where $\theta = (\text{shift, squeeze})$ enters the columns of Φ through the resampling of the reference cross sections, and c collects the gas amounts and the Chebyshev baseline coefficients. The problem is separable: for any θ the linear parameters are determined by least squares, $c^*(\theta) = \Phi^+ \alpha$, and the objective reduces to the *profiled* residual

$$r(\theta) = (I - \Phi(\theta)\Phi(\theta)^+) \alpha = P^\perp(\theta) \alpha.$$

The optimiser searches only θ ; c is never a search direction. This is variable projection (Golub and Pereyra, 1973).

Table B1. Effect of rescaling one column of the small test problem of Appendix B2 on the condition number κ , on the projector $P^\perp$ and on the derivative $\partial r / \partial \theta$.

column scaling	$\kappa(\Phi D)$	$\ \Delta P^\perp\ $	relative change in $\partial r / \partial \theta$
none	9.3	0	0
10^{-3} on one column	6.1×10^5	2.4×10^{-10}	1.1×10^{-10}
10^{-8}	6.1×10^{11}	1.3×10^{-4}	5.1×10^{-5}
10^{-12}	7.7×10^{15}	1.00	0.48
10^{-19}	2.1×10^{16}	1.00	0.48

B2 The derivative has two terms

Differentiating the projector rather than the fit gives, for each nonlinear parameter θ_j and with $\Phi_j = \partial \Phi / \partial \theta_j$,

$$\frac{\partial r}{\partial \theta_j} = - \left[P^\perp \Phi_j \Phi^+ + (P^\perp \Phi_j \Phi^+)^T \right] \alpha.$$

The first term is the change in the residual caused by moving the column space; the second is the change caused by the reprojection of α onto it. Kaufman (1975) drops the second, which is cheaper and is often adequate because it does not change the stationary point – both expressions vanish where the true gradient vanishes – only the path taken to it and the curvature reported there.

It is not adequate here. On a small separable problem with the same structure as Sect. 2.1 – two Gaussian absorber columns and three Chebyshev columns – the two-term expression agrees with a central finite difference of $r(\theta)$ to 2×10^{-10} relative, while the one-term expression is in error by 20 % and 17 % in the two nonlinear parameters. The step-size sweep of Sect. 3.2 measures the same thing on the real problem: the analytic derivative descends to the round-off floor, which an expression missing a term of that size could not do.

B3 Column scaling changes nothing, until it changes everything

Replace Φ by ΦD for any invertible diagonal D . Then $(\Phi D)^+ = D^{-1} \Phi^+$, so $P = \Phi D D^{-1} \Phi^+ = \Phi \Phi^+$ and $P^\perp$ are unchanged, and in the derivative of Appendix B2 the factors D and D^{-1} cancel term by term. The profiled residual, its derivative and the fitted θ are therefore invariant under rescaling of the columns; only c and its covariance rescale, and they rescale exactly inversely, so the retrieved amounts are unchanged.

That is the algebra. The arithmetic is not invariant, and the failure is abrupt rather than gradual, because Φ^+ is computed with a rank tolerance (Table B1).

Once the scaling drives the condition number past the tolerance the pseudo-inverse discards the column, the projector is onto a different subspace, and the invariance is lost completely rather than degraded. The operational problem sits at the bottom of that table by construction: a cross-section column is of order 10^{-19} cm^2 and a Chebyshev column is of order unity. Column

normalisation is therefore not a numerical nicety but the condition under which the invariance of Appendix B3 holds at all, and the retrieval normalises every column to unit norm before the solve.

990 **B4 The joint covariance and what the fit usually reports**

Linearising the full problem in (c, θ) at the solution gives the Jacobian $J = [\Phi \ B]$ with $B = \partial(\Phi c)/\partial\theta$, and the covariance $\sigma^2(J^\top J)^{-1}$. Partitioning it, the θ block is the Schur complement

$$\text{Cov}(\theta) = \sigma^2(B^\top P^\perp B)^{-1},$$

which is the projected curvature of Appendix B2 and not $\sigma^2(B^\top B)^{-1}$: the part of B that lies in the column space of Φ is
 995 absorbed by the linear parameters and constrains θ not at all. The c block is correspondingly larger than the $(\Phi^\top \Phi)^{-1}$ that a fit with θ held fixed would return, by an amount that depends on how much of each column's sensitivity the nonlinear parameters can mimic. On the toy problem above the inflation runs from 1.001 to 1.257 across the five columns.

Both statements are elementary and neither is new. What matters for Sect. 5 is that a retrieval which reports $\sigma^2(\Phi^\top \Phi)^{-1}$ – the covariance of the linear parameters at a fixed θ – is reporting a quantity that is correct only if θ was known rather than
 1000 fitted, and the difference is not small when θ is poorly determined. Augur forms the full $(J^\top J)^{-1}$ and reports both blocks; the θ block is available at no extra cost once the projected Jacobian of Appendix B2 has been formed. It describes only the local curvature at the solution, which is why the identifiability statistics of Sect. 5 are taken from profile scans of the projected objective rather than from this block.

Appendix C: Calibration-block length and duty cycle

1005 This appendix gives the measurements behind the second paragraph of Sect. 6.2. They are not conclusive, for the reason stated below, and are recorded so that the measurement that would settle the question can be designed.

What acts on knot noise is the number of scans averaged into each calibration block. Splitting each block into contiguous quarters separates the two components that limit a block mean – white scan noise σ_s , which averages down, and drift within the block at rate g , which does not:

$$\begin{aligned} \text{scale}(Q_0 - Q_1)^2 &= \frac{8\sigma_s^2}{N} + \left(\frac{gT}{4}\right)^2, \\ 1010 \quad \text{scale}\left(\frac{Q_0+Q_1}{2} - \frac{Q_2+Q_3}{2}\right)^2 &= \frac{4\sigma_s^2}{N} + \left(\frac{gT}{2}\right)^2. \end{aligned} \tag{C1}$$

Sub-blocks must be cut contiguously for this, because an interleaved (even/odd) split distributes neighbouring scans across both halves and cancels whatever they share. The interleaved split is kept as a control: the ratio between the two, at the same number of scans, measures how much of the block's scatter is common to neighbouring scans rather than independent.

The operational block is $N = 34$ scans spanning $T = 32.0$ s, in both channels and for both reference gases. Within it, aver-
 1015 aging does not behave as white noise (Table C1).

Table C1. Scatter of contiguous sub-block contrasts within the operational calibration block ($N = 34, T = 32.0$ s) in the two heated channels, the averaging exponent they imply, and the ratio of contiguous to interleaved scatter at the same number of scans m , with the values expected for white noise.

	hot ANs	hot PNs	white noise
contiguous quarters (8 scans)	2.506×10^{-4}	1.095×10^{-4}	
contiguous halves (16 scans)	2.332×10^{-4}	0.820×10^{-4}	
averaging exponent	0.104	0.416	0.500
contiguous / interleaved, same m	2.26	1.22	1.03

Table C2. Calibration duty cycle for the operational zero-air block and for blocks two and four times longer.

zero-air block	N	zero-air duty	total calibration duty
operational	34	0.90 %	1.20 %
$\times 2$	68	1.79 %	2.10 %
$\times 4$	136	3.58 %	3.89 %

Doubling the number of scans averaged reduces the scatter by 7 % in the ANs channel, against the 29 % that independent scans would give. Most of what a block mean contains is therefore not independent between scans, and the last row says so directly: cutting the same block contiguously rather than in alternation more than doubles its apparent scatter in ANs. These are spectral-domain statistics: in the NO₂ amount, contiguous and interleaved halves give similar knot noise (Sect. 4.2), so most of this correlated structure is evidently absorbed by the baseline polynomial before it reaches the gas columns. A monotonic trend across the 32 s is present in both channels but explains little of this – see below.

Two conclusions follow, and the second is a limitation. Within the range that can be measured, lengthening the averaging window is worth much less than the $N^{-1/2}$ a reader would assume, and how much less depends on the channel. And the range that can be measured is entirely below the operational block length: subdividing N scans says nothing about $2N$ or $4N$. The evidence available is therefore not neutral but mildly unfavourable, and the measurement that would settle it – operating one longer block – has not been made.

The duty cycle is not what stands in the way. Calibration occupies 1.2 % of the measurement time (Sect. 6.4), and a fourfold zero-air block would take it to 3.9 % (Table C2).

Even under the white-noise assumption the return would be modest, and it is worth stating why before anyone spends duty cycle on it. Halving the zero-air knot noise removes three quarters of the I_0 variance, but that term carries 45 % of the Σ ANs variance, so the structural budget would fall from 0.1236 to 0.1008 ppb – 18.5 %, not half. The measured exponents put even that out of reach in one channel; Table C3 gives the ceiling for each remedy.

These are ceilings, not forecasts, and the way they are computed needs stating because Sect. 4.3 forbids the obvious method. Removing a term’s contribution from a total is a quadrature operation, and the joint propagation is not a quadrature sum –

Table C3. Best attainable Σ ANs structural uncertainty (standard deviation) under each remedy, as a fraction of the operational value, and the assumption each row rests on. These are ceilings, not forecasts.

	Σ ANs structural (SD), best attainable	of operational	status
operational	0.1236 ppb	1.00	measured
$\times 4$ blocks only	0.1008	0.82	assumes $N^{-1/2}$; not supported below N
flag bad R intervals only	0.0901	0.73	assumes a perfect detector
both	0.0547	0.44	both assumptions

1035 here it exceeds one by 7.6 %, 0.1236 ppb against 0.1149. The rows are therefore evaluated in quadrature space and rescaled by that measured ratio, which makes the operational row reproduce the measured joint exactly and applies the same correction to the others. The residual assumption is that the ratio survives shrinking a term, which it will not do exactly, because the ratio depends on the covariance and interaction of the terms (Sect. 4.3), and shrinking one term changes them.

1040 The ratio column is unaffected by any of this: the rescaling factor cancels, so the relative ordering is the same whether the calculation is done in quadrature space or after rescaling. Section 6.2 uses only the ordering.

Flagging outranks averaging here, and it does so under every basis we have checked – record space as well as hourly, and quadrature space as well as rescaled. The flagging row assumes a detector that removes the tail and nothing else, reducing the R term to its present robust scale; a real flag will fall short. The block row assumes an averaging behaviour the measurements of this appendix do not support.

1045 A linear trend across the block is part of the explanation but not all of it. It accounts for 40 % of the within-block variance in the ANs channel and 34 % in PNs, inflating the apparent sub-block scatter by 1.29 and 1.23. Whether removing it restores white averaging cannot be tested by detrending and re-measuring: the contrast comparing contiguous halves and the least-squares slope over four sub-blocks have a cosine of -0.894 , so subtracting the line subtracts most of the quantity being measured. Measured instead on a contrast orthogonal to the trend, the residual scatter is still 1.88 times white in ANs and 1.17 in PNs.

1050 Detrending is a partial remedy; what limits averaging is a correlation that outlives the removal of its linear part.

What does not depend on either assumption is the ordering. The reflectivity term is concentrated in a few episodes and does not average down; the zero-air term is distributed and averages at least partly. Flagging comes first whatever the detector achieves and whatever the block scaling turns out to be.

1055 One observation from the same measurement is worth recording. Both channels show a monotonic trend across the 32 s block, of opposite sign $- +248$ to -314 ppm in ANs against -261 to $+235$ ppm in PNs. Two adjacent windows moving in opposite directions is a change of spectral slope rather than of intensity, which points at the source or at the bandpass separating the channels rather than at the gas fill. Removing the linear trend by contrast does not reduce the excess scatter, so this trend is not what limits averaging; the correlation between neighbouring scans is.

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1060 *Competing interests.* To be completed by the authors.

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